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Faculty of Computers and Information

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CareerCompass: AI-Powered Career Guidance and Job Recommendation Platform

CareerCompass

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Acknowledgment

The project team would like to express sincere appreciation to Dr. Amna Mahmoud for academic supervision, technical guidance, and continuous feedback during the preparation of CareerCompass. The team also thanks the Faculty of Computers and Information at Kafr El-Sheikh University for providing the academic setting in which this graduation project was designed, implemented, tested, and documented.

The work presented in this book reflects a collaborative software engineering effort. It combines web application development, database design, AI-assisted document analysis, explainable matching, containerized deployment, testing, and technical documentation. The two supervisor-provided graduation books were used only to understand expected report structure and visual formality; no content, wording, project-specific claims, diagrams, or references were copied from them.

Abstract

CareerCompass is a graduation/demo career guidance platform that helps students and early-career users understand their CV profile, explore imported job opportunities, and compare their current skills against job requirements. The system consists of a React and Vite frontend, a Laravel API backend, a MySQL database, a FastAPI-based CV analyzer, a FastAPI/Scrapy-based job miner, MinIO-compatible private file storage, Nginx routing, and Prometheus/Grafana monitoring. The platform supports registration, login, CV upload, AI-assisted CV parsing, normalized profile and skills storage, job recommendation, gap analysis, an application tracker, and administrator dashboards for job and source diagnostics.

The implementation is intentionally described as a graduation/demo system rather than a production product. The AI outputs are estimates, the job data depends on imported and demo sources, and the security posture is appropriate for demonstration but requires further production hardening. Validation was performed through Docker Compose configuration checks, backend tests, frontend lint/build, Python service tests or syntax checks, HTTP probes, and manual browser screenshots. Backend tests passed with 39 tests and 297 assertions, the AI job miner tests passed with 75 tests, and the frontend build completed successfully. The AI CV analyzer container did not include pytest, so its pytest suite was marked as skipped while Python syntax compilation passed.

Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CV	Curriculum Vitae
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
RBAC	Role-Based Access Control
REST	Representational State Transfer
S3	Simple Storage Service compatible object storage
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface

Chapter 1: Introduction

1.1 Introduction to the Project

CareerCompass is an AI-assisted career guidance and job recommendation platform developed as a Computer Science graduation project for Kafr El-Sheikh University. The system helps a student upload a CV, receive a structured profile, view estimated job matches, inspect skill gaps, and save opportunities in an application tracker. The project combines web engineering, backend service design, natural language processing support, web scraping support, data persistence, and containerized operations.

The platform uses a modern web stack. The frontend is implemented with React, a component-based UI library [4], and bundled with Vite [5]. The backend uses Laravel, which provides routing, controllers, validation, middleware, Eloquent ORM, queues, testing support, and file storage abstractions [1]. The AI services are implemented with FastAPI, a Python API framework that supports type-hinted request and response handling [6]. The deployment is orchestrated through Docker Compose [11].

1.2 Background and Motivation

Many students prepare CVs and search for jobs without a clear view of how their current skills relate to job descriptions. Career guidance is often fragmented across CV feedback, job boards, learning resources, and manual advice. CareerCompass addresses this academic problem by placing those steps into one integrated demonstration system. The objective is not to replace human career advising, but to provide a practical software prototype that can parse a CV, normalize profile data, import job records, estimate matches, and present explainable gap analysis.

1.3 Problem Statement

Students frequently face three connected problems. First, they may not know whether their CV communicates their skills clearly. Second, job listings often describe requirements in different formats, making comparison difficult. Third, students may save opportunities across separate tools without a coherent tracker. CareerCompass proposes a centralized graduation/demo system that links CV data, jobs, matching, gap analysis, and tracking.

1.4 Purpose of the Project

The purpose of CareerCompass is to demonstrate how a distributed web system can support career exploration using explainable AI-assisted workflows. The project demonstrates authentication, CV upload, private storage, parsing, skill extraction, job import, recommendation, gap analysis, admin diagnostics, and monitoring in a Dockerized environment.

1.5 Project Objectives

- Provide a student-facing web interface for account creation, login, CV upload, profile review, job recommendations, gap analysis, and application tracking.
- Provide an administrator interface for dashboard statistics, job review, source diagnostics, and target role management.
- Implement a Laravel API that protects authenticated workflows using token-based access and role checks.
- Implement Python services for CV analysis and job mining support.
- Store uploaded CV files privately and expose downloads through controlled URLs.
- Validate the system with backend tests, frontend lint/build, Python tests where available, Docker checks, HTTP probes, and browser screenshots.

1.6 Proposed Solution

The proposed solution is a multi-service application. React renders the browser interface. Nginx serves as the gateway. Laravel handles REST-style APIs over HTTP [16], authentication, validation, data models, business services, and admin routes. MySQL stores normalized data [8]. MinIO provides S3-compatible object storage [9]. The CV analyzer parses PDFs and images using Python text extraction and OCR-related libraries. The job miner imports jobs from demo, API, and scraping-style sources using FastAPI, Scrapy, and HTML parsing concepts [16], [17], [18].

1.7 Project Scope

The scope covers a graduation/demo environment: local Docker deployment, student workflows, admin workflows, testing, screenshots, and documentation. It does not claim production readiness, job placement certainty, full job market coverage, legal compliance for production privacy, or complete AI accuracy.

1.8 Graduation Demo Positioning

CareerCompass should be presented as a working academic prototype with realistic engineering boundaries. The screenshots and tests show that the core workflows can run locally. However, large-scale evaluation, production identity management, cloud infrastructure, observability hardening, legal privacy review, and complete market integrations remain future work.

1.9 Report Organization

Chapter 2 analyzes requirements and users. Chapter 3 presents architecture, diagrams, database design, and deployment. Chapter 4 lists software and tools with references. Chapter 5 documents implementation modules from the repository. Chapter 6 presents testing and evaluation results. Chapter 7 discusses security and privacy. Chapter 8 concludes with achievements, limitations, and future work.

Chapter 2: System Analysis

2.1 Introduction

System analysis defines what CareerCompass should do, who uses it, and what constraints influence implementation. The analysis is based on the reviewed repository, including the root documentation, Docker configuration, Laravel API, React frontend, Python services, database migrations, seeders, tests, and finalization notes.

2.2 Development Methodology

The project followed an iterative engineering approach. Each feature was implemented, checked through local commands or tests, integrated with Docker services, then documented. This approach is appropriate for a graduation project because it supports incremental progress while keeping the system demonstrable. The architecture uses separate services, which follows a common microservice-style pattern where independently deployable services communicate over network APIs [29].

2.3 Existing Career Guidance Problems

The system targets common problems in student career preparation:

- CV content is not structured for direct software comparison.
- Job descriptions vary in wording and completeness.
- Students need explainable feedback rather than unexplained match numbers.
- Admin users need visibility into imported jobs and scraping sources.
- Demo systems need reliable startup, testing, and evidence for academic evaluation.

2.4 Target Users and Stakeholders

Stakeholder	Description	Main Interest
Student user	A university student or early-career user.	Upload CV, view profile, discover jobs, analyze gaps, save opportunities.
Administrator	A project operator or supervisor/demo administrator.	Review jobs, source diagnostics, users, target roles, and system health.
Supervisor	Academic supervisor evaluating the graduation project.	Correctness, completeness, originality, and honest evaluation.
Project team	Developers responsible for design and implementation.	Maintainable code, demonstrable workflows, testing, and documentation.

Table 1. Stakeholder summary.

2.5 User Roles

CareerCompass implements two practical roles. The student role can register, login, upload a CV, view recommendations, run gap analysis, and track applications. The admin role can access protected admin routes for dashboard statistics, job administration, scraping sources, target roles, and user review.

2.6 Functional Requirements

ID	Requirement	Implementation Evidence
FR-01	Register and login users.	Laravel AuthController, RegisterRequest, LoginRequest, React Login/Register pages.

ID	Requirement	Implementation Evidence
FR-02	Upload a CV file.	CvUploadRequest validates PDF/JPEG/PNG and max size; Dashboard appends file field cv.
FR-03	Store CV files privately.	CvStorageService and MinIO/S3-compatible disk configuration.
FR-04	Parse CV and extract profile/skills.	CvProcessingService and AI CV Analyzer /api/parse-cv.
FR-05	Display normalized profile and skills.	UserResource, Profile page, Dashboard AI insight components.
FR-06	Import and display jobs.	JobPosting model, ScrapedJobController, JobController, AI Job Miner.
FR-07	Estimate job recommendations.	JobController and GapAnalysisService use CV/profile data and matching service.
FR-08	Analyze skill gaps.	GapAnalysisController and GapAnalysis page.
FR-09	Track applications.	ApplicationController, ApplicationTrackerService, Applications page.
FR-10	Provide admin dashboards.	Admin Dashboard, Jobs, Sources, Targets pages and admin API routes.
FR-11	Provide health and metrics endpoints.	HealthController, MetricsController, Prometheus/Grafana compose services.

Table 2. Functional requirements summary.

2.7 Non-Functional Requirements

Category	Requirement	CareerCompass Approach
Usability	The UI should guide students through CV upload and recommendations.	React pages, dashboard cards, profile score, and action buttons.
Maintainability	Code should be modular.	Controllers, requests, services, resources, models, React pages, and Python services are separated.
Reliability	The system should degrade honestly if AI services fail.	CV processing returns timeout/error/no_text statuses and preserves prior data when appropriate.
Security	Authentication, validation, private files, and admin role checks are required.	Sanctum tokens, request validation, admin middleware, signed URLs, service tokens.
Observability	Health and metrics should be available.	/api/health, /api/ready, /api/metrics, Prometheus, Grafana.
Portability	The demo should run locally.	Docker Compose services and environment examples.

Table 3. Non-functional requirements summary.

2.8 Hardware Requirements

For local demonstration, a developer machine capable of running Docker Desktop and multiple containers is required. CV parsing and OCR-like processing can be CPU-intensive; therefore, enough memory should be available for the Laravel backend, MySQL, frontend, Python services, MinIO, Prometheus, and Grafana. GPU acceleration is not required for the demonstrated flow.

2.9 Software Requirements

Layer	Software
Frontend	React, Vite, Tailwind-style CSS classes, lucide-react icons, Recharts.
Backend	Laravel 12, PHP, Composer dependencies, Sanctum-style token authentication.
AI services	Python, FastAPI, text extraction, OCR-related dependencies, Scrapy-style job mining.
Data	MySQL 8.0, MinIO/S3-compatible storage.
Infrastructure	Docker, Docker Compose, Nginx, Prometheus, Grafana.
Testing	PHPUnit/Pest-style Laravel tests, pytest for Python, ESLint and Vite build.

Table 4. Hardware and software environment.

2.10 Input and Output Flow

Primary inputs include user account data, uploaded CV files, imported job records, target role settings, and administrator source configurations. Primary outputs include normalized user profiles, skill lists, CV analysis metadata, estimated job matches, gap reports, application records, admin statistics, health checks, and monitoring metrics.

2.11 Use Case Summary

The main use cases are shown in Figure 5. The system separates student and administrator responsibilities while sharing the same backend API and database.

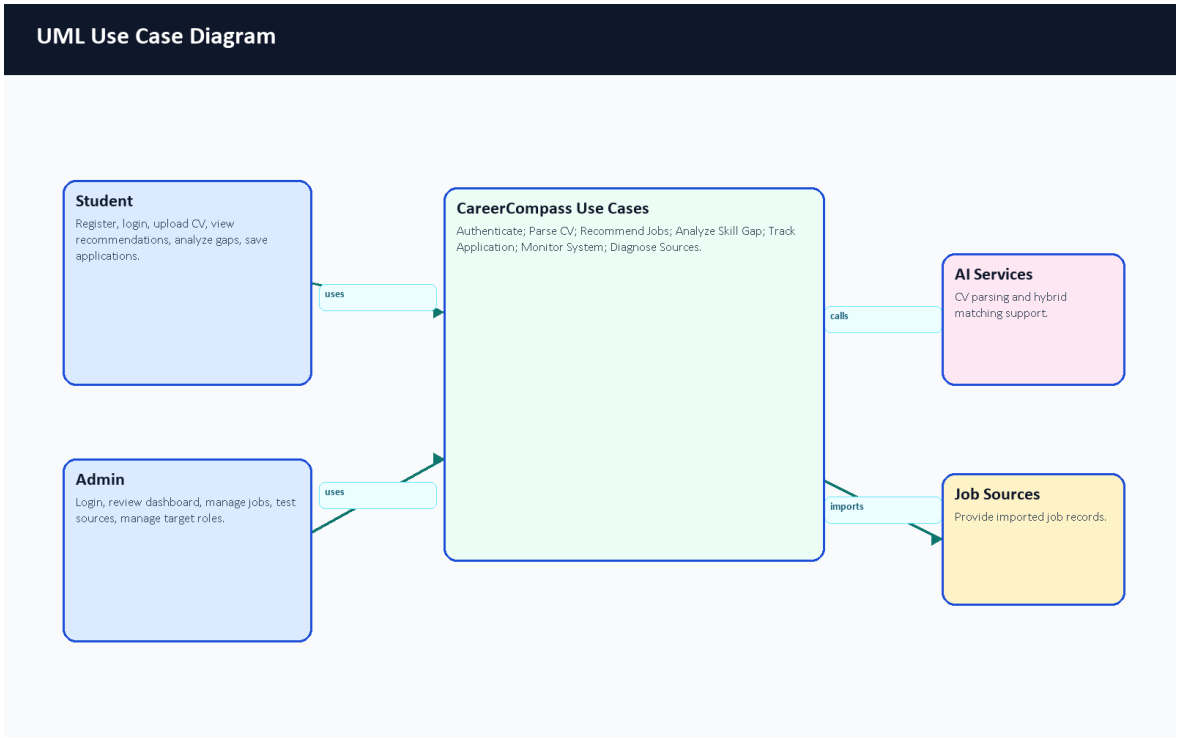


Figure 5. UML use case diagram.

Chapter 3: System Design and Architecture

3.1 Introduction

CareerCompass is designed as a Dockerized multi-service application. This design separates browser UI, API logic, AI services, data storage, object storage, reverse proxy routing, and monitoring. Docker containers help package runtime dependencies consistently [10], while Docker Compose coordinates the multi-container local deployment [11].

3.2 High-Level System Architecture

The high-level architecture is shown in Figure 1. Browser users interact with the React frontend through Nginx. The frontend calls the Laravel API. Laravel persists records in MySQL, stores CV files in MinIO-compatible storage, calls the AI CV Analyzer for parsing, calls matching logic for recommendations/gaps, and receives job imports from the job miner.

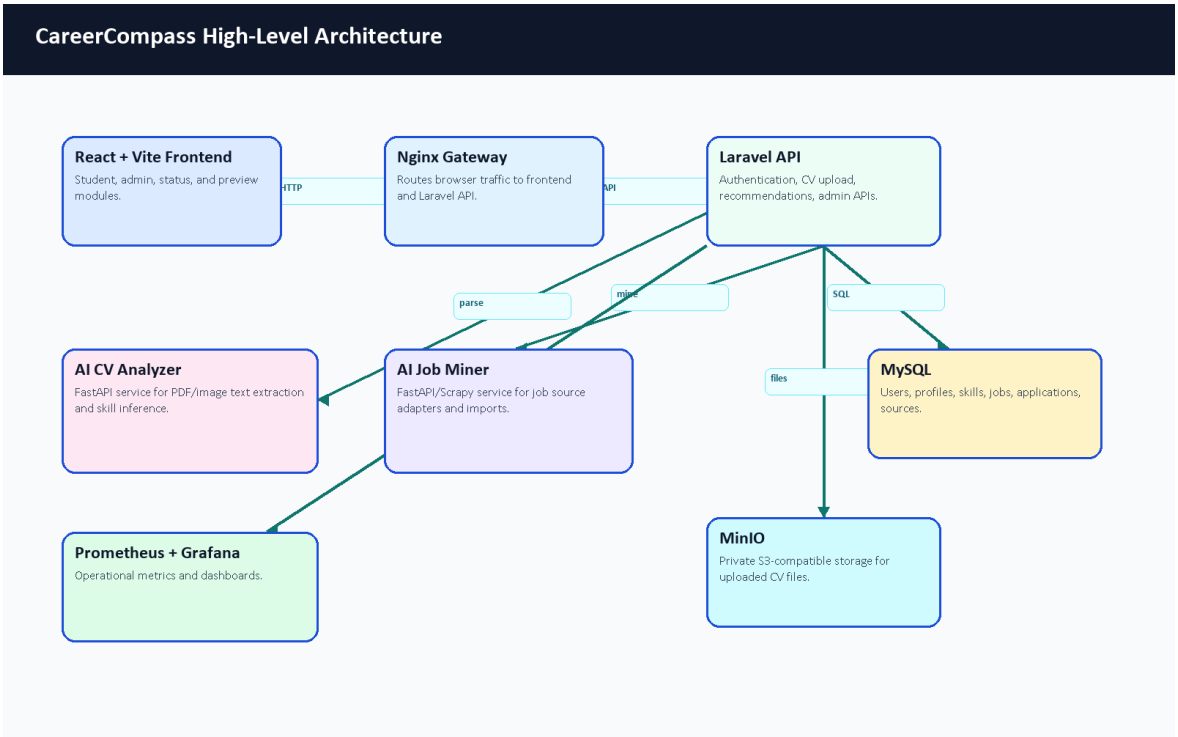


Figure 1. High-level architecture of CareerCompass.

3.3 Frontend Architecture

The frontend is implemented with React and organized around routes in frontend/src/App.jsx. Public pages include Home, Login, Register, About, Privacy, Terms, and System Status. Protected student routes include Dashboard, Jobs, Gap Analysis, Profile, Settings, Market Intelligence, Applications, CV Builder, Mock Interview, Learning, Career Planner, Mentorship, and Tools Hub. Protected admin routes include Admin Dashboard, Jobs, Users, Sources, and Target Roles.

The frontend API layer is located under frontend/src/api. Axios is configured in client.js, including base URL resolution, bearer token injection, request IDs, retry behavior for safe GET requests, and 401 handling. Authentication state is managed by AuthContext.jsx, which stores the user and token in local storage. Localization files exist under frontend/src/locales.

3.4 Backend API Architecture

The backend is a Laravel API. Routes are defined in backend-api/routes/api.php and are registered both at /api and /api/v1. The API includes public health/readiness/metrics endpoints, guest authentication routes, public job listing routes, internal scraper import routes protected by a service token, authenticated student routes, and admin routes protected by middleware.

Laravel provides structured controllers, form requests, resources, services, models, migrations, seeders, and tests. This aligns with Laravel's documented framework responsibilities, including routing, validation, database access, queues, and testing [1], [3].

3.5 AI CV Analyzer Architecture

The AI CV Analyzer is a FastAPI service. Laravel sends CV files to this service for parsing. The analyzer extracts readable text from PDF or image inputs, attempts structured inference, and returns fields such as predicted role, seniority, domain, skills, strengths, gaps, red flags, confidence, and parsing status. The backend handles statuses such as success, OCR fallback, timeout, error, empty file, and no text. PDF and OCR-related libraries are supported by external tools such as PyMuPDF, pdfplumber, and EasyOCR [22], [23], [24].

3.6 AI Job Miner Architecture

The AI Job Miner is a FastAPI service with scraping and import support. It includes source adapters for demo/local data, public APIs, and HTML/Scrapy-style extraction. Scrapy is a Python framework for extracting structured data from websites [17], and Beautiful Soup is commonly used to parse HTML documents [18]. CareerCompass uses a quality gate and honest source classifications so that demo/imported data is not overstated as complete market coverage.

3.7 Database Design

MySQL stores users, profiles, skills, experience records, CV analyses, job postings, applications, scraping sources, target job roles, scraping jobs, failed URLs, and related metadata. MySQL is a relational database system documented by Oracle [8]. The Laravel migrations define schema constraints, indexes, foreign keys, and unique combinations such as job title/company uniqueness.

3.8 ERD

Figure 8 summarizes the main database tables and relationships. It is not a complete replacement for migrations, but it provides a readable graduation-book view of the data model.

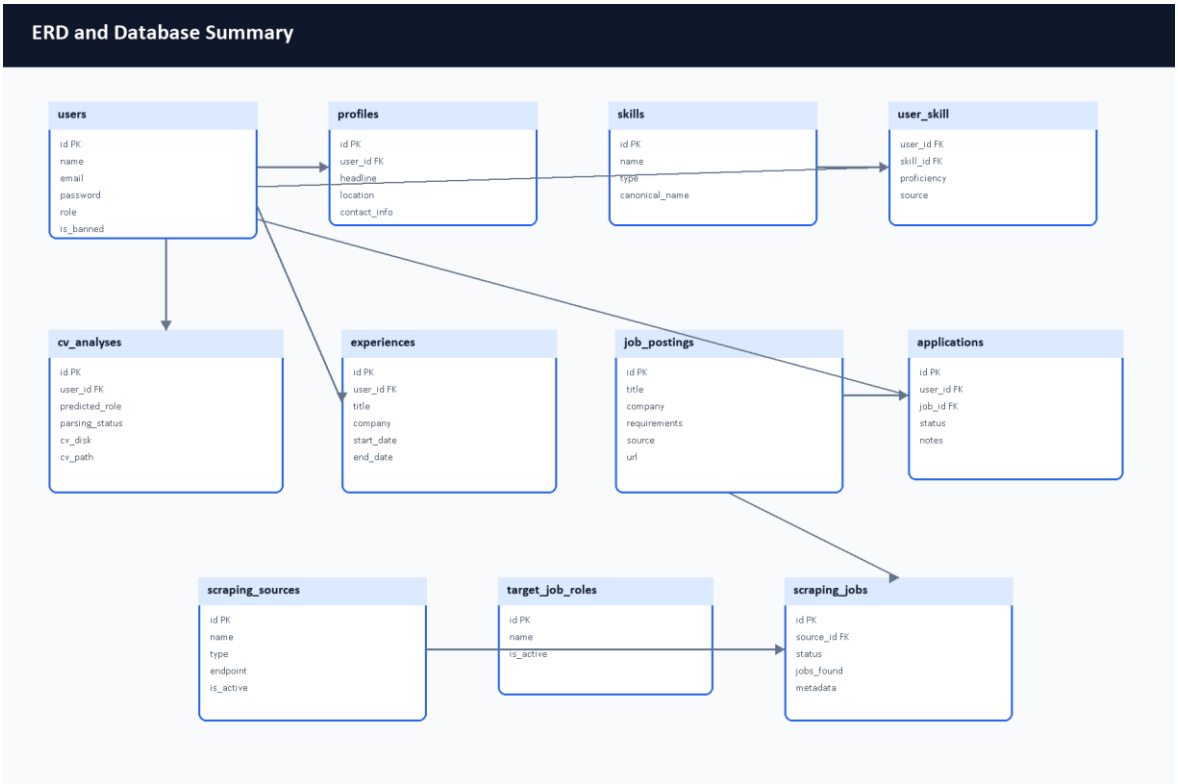


Figure 8. ERD and database summary diagram.

3.9 Data Flow Diagrams

The context-level data flow is shown in Figure 3, and the expanded process-level view is shown in Figure 4. Student and administrator workflows enter the same system boundary, while external job sources and AI services interact with controlled backend processes.

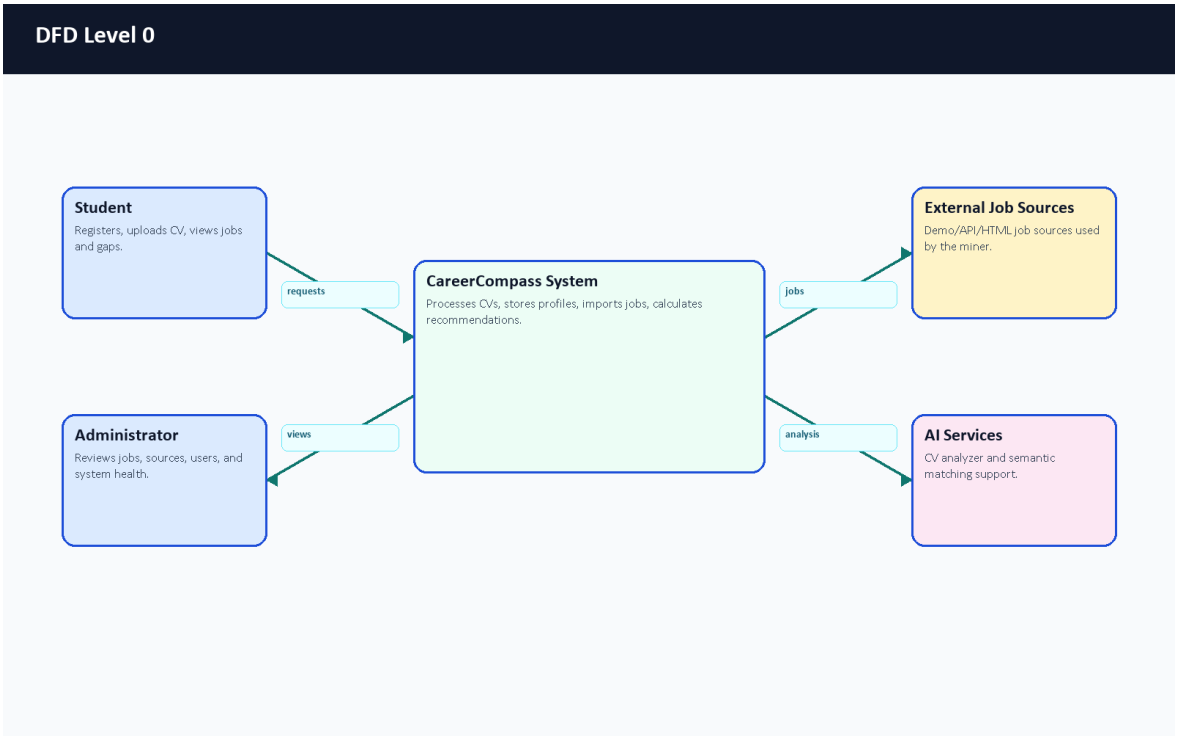


Figure 3. DFD Level 0 context diagram.

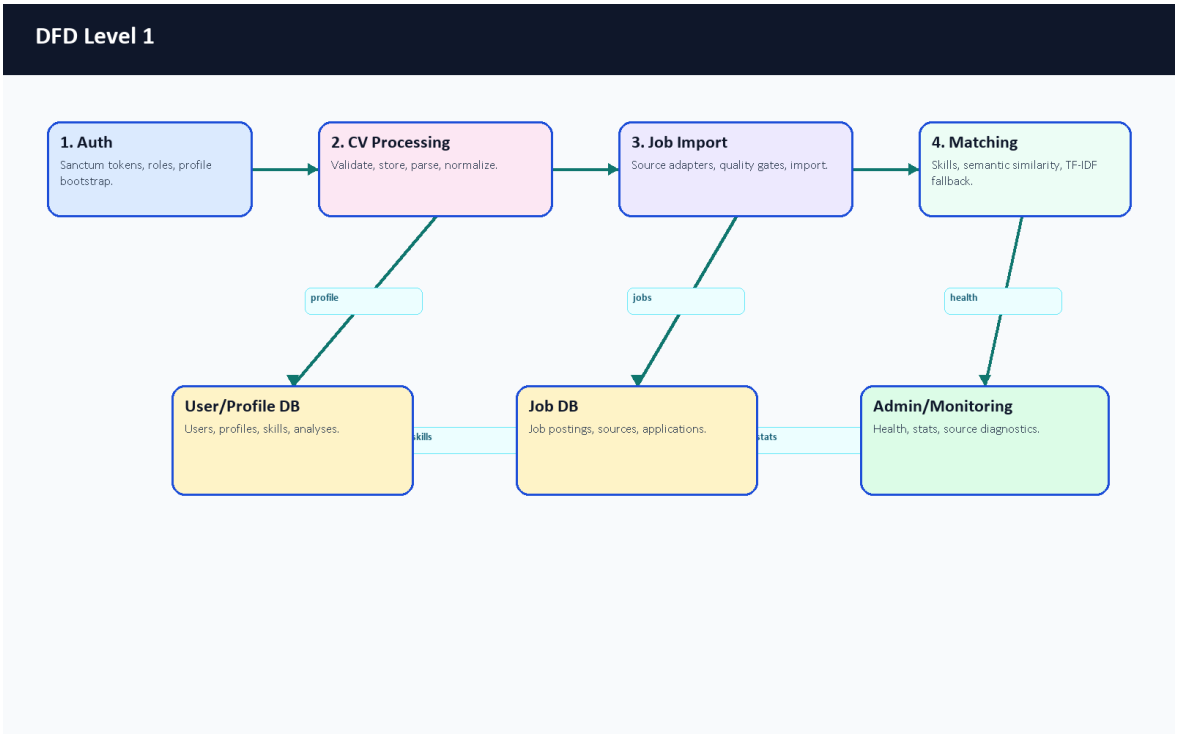


Figure 4. DFD Level 1 process diagram.

3.10 UML Use Case Diagram

The use case diagram separates student actions from administrator actions. Student workflows focus on career exploration. Admin workflows focus on operating and inspecting the imported job ecosystem.

3.11 UML Sequence Diagrams

Figure 6 shows the CV upload and analysis sequence. Figure 7 shows recommendation and gap analysis.

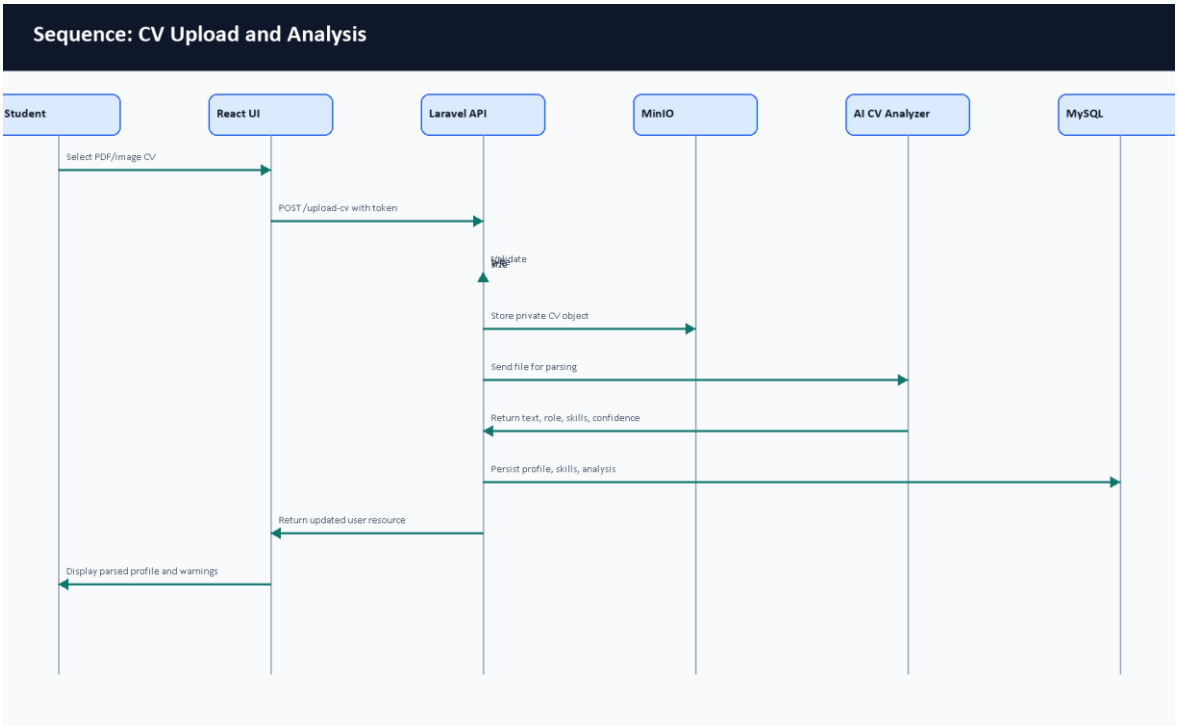


Figure 6. Sequence diagram for CV upload and analysis.

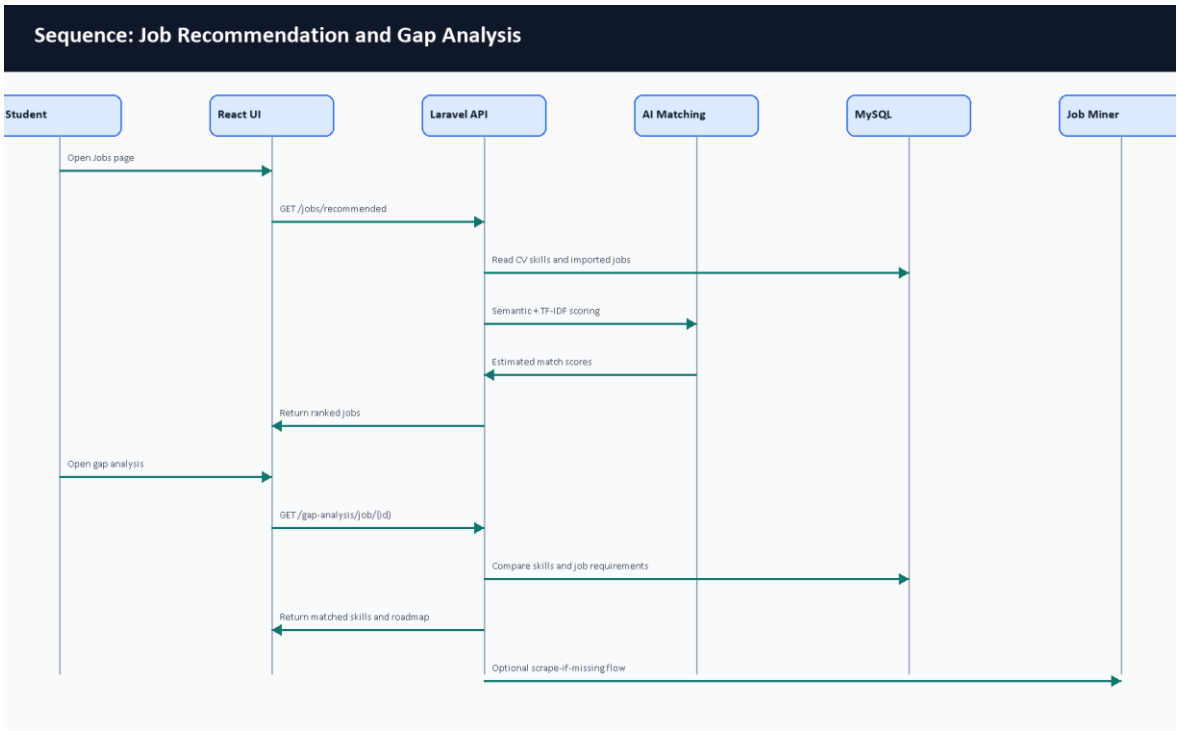


Figure 7. Sequence diagram for recommendation and gap analysis.

3.12 Deployment Architecture with Docker

The deployment is defined by Docker Compose files. Nginx exposes the application, the frontend serves built React assets, the Laravel API and workers handle backend work, MySQL stores structured data, MinIO stores private CV objects, Python services provide AI CV parsing and job mining, and Prometheus/Grafana provide monitoring. Figure 2 summarizes the container layout.

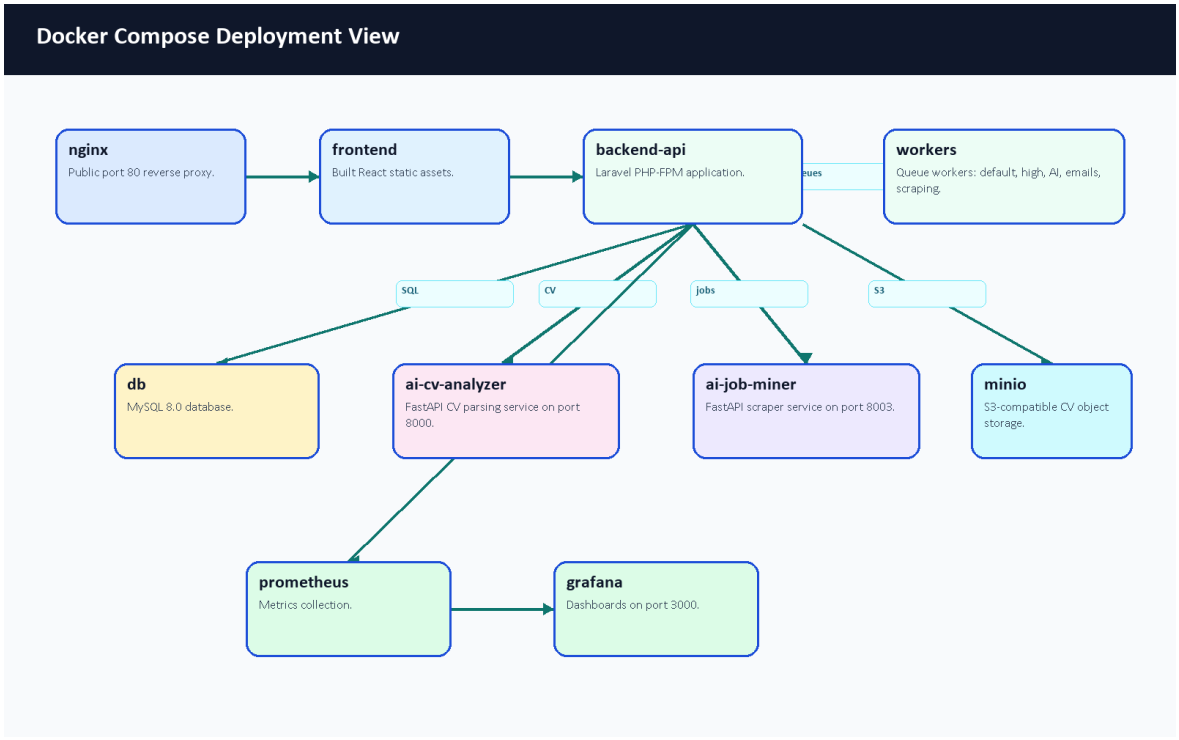


Figure 2. Docker deployment architecture.

3.13 Monitoring Architecture

CareerCompass includes live, readiness, and metrics endpoints. Prometheus is used for scraping and time-series metrics collection [13], while Grafana visualizes metrics and dashboard panels [14]. The admin dashboard also exposes application-level health information for the graduation demo.

3.14 Design Decisions and Justification

Decision	Justification
Use Laravel for the main API.	The project benefits from built-in routing, validation, Eloquent ORM, queues, resources, tests, and middleware [1].
Use React for UI.	Component-based pages support student/admin route separation and reusable cards [4].
Use FastAPI for AI services.	Python AI/NLP dependencies are easier to isolate behind typed HTTP services [6].
Use Docker Compose.	Multiple services can be started consistently for a graduation defense [11].
Use private object storage for CV files.	CV files are sensitive; private storage and signed downloads reduce accidental exposure [9], [27].
Keep AI wording honest.	Match scores and CV parsing are estimates; the demo should avoid claiming certain outcomes.

Table 5. Design decisions summary.

Chapter 4: Software and Tools Used

4.1 Laravel

Laravel is used for the backend API, controllers, middleware, validation requests, Eloquent models, migrations, resources, queues, tests, and storage integration [1]. In CareerCompass, Laravel centralizes authenticated business logic and data persistence.

4.2 React

React is used to build the browser-based interface for students and administrators [4]. The project uses React routes and components for dashboard cards, forms, pages, tables, modals, and visualization panels.

4.3 Vite

Vite is used as the frontend build tool [5]. The successful production build transformed 2904 modules during validation.

4.4 FastAPI

FastAPI is used for Python services because it supports efficient API development using Python type hints and modern web service patterns [6]. CareerCompass uses it for CV analysis and job mining service endpoints.

4.5 Python

Python is used for AI and scraping-related services [7]. It supports the ecosystem of text extraction, OCR, NLP, testing, and scraping libraries used by the AI services.

4.6 MySQL

MySQL stores relational project data including users, profiles, skills, jobs, applications, scraping sources, and CV analyses [8].

4.7 MinIO / S3-Compatible Storage

MinIO-compatible object storage is used to store CV files privately. This avoids placing uploaded CVs directly in public web directories and supports signed temporary download flows [9].

4.8 Docker and Docker Compose

Docker containers package each runtime, and Docker Compose coordinates multi-container execution [10], [11]. CareerCompass uses Compose for Nginx, frontend, backend, workers, MySQL, MinIO, AI services, Prometheus, and Grafana.

4.9 Nginx

Nginx acts as the reverse proxy and public gateway for the local deployment [12]. It routes browser and API traffic to the correct containers.

4.10 Prometheus and Grafana

Prometheus collects metrics [13], and Grafana visualizes metrics and dashboards [14]. CareerCompass uses these tools to support monitoring-oriented evaluation.

4.11 GitHub Actions

GitHub Actions is configured for repository automation and CI/CD-style validation [15]. The report treats workflow definitions as part of the development process, while final PR status should be reviewed after the draft PR is opened.

4.12 NLP / AI Libraries

The project uses concepts and libraries related to text extraction, OCR, TF-IDF, cosine similarity, and sentence embeddings. TF-IDF and cosine similarity are documented by scikit-learn [19], [20]. Sentence Transformers provides sentence embedding models and utilities [21]. PyMuPDF, pdfplumber, and EasyOCR support PDF/image text extraction and OCR-style workflows [22], [23], [24].

4.13 Testing Tools

Backend tests use Laravel/PHP testing tools and PHPUnit concepts [25]. Python service tests use pytest where available [26]. Frontend validation uses ESLint and Vite build checks.

4.14 Development and Version Control Tools

Git and GitHub are used for version control and pull-request-based collaboration. Docker Desktop provides the local container runtime. Browser screenshots were captured through a Chrome DevTools Protocol helper to provide evidence images for this report.

Chapter 5: System Implementation

5.1 Introduction

This chapter documents the implemented CareerCompass modules based on repository files. The implementation is not a generic career platform; it is a Laravel/React/FastAPI/Docker system with specific routes, services, pages, models, seeders, and tests.

5.2 Authentication and User Management

Authentication is implemented in backend-api/app/Http/Controllers/Api/AuthController.php. Registration creates a user with role user, loads profile and related resources, and returns a token. Login validates credentials, checks the banned state, revokes old tokens, creates a new token, and returns a user resource. The frontend stores the token as auth_token and the user object in local storage through frontend/src/context/AuthContext.jsx.

The register request restricts emails to selected public email domains and validates password format. The admin role is protected by the IsAdmin middleware. Admin seed data creates a demo-only administrator account through AdminUserSeeder.

5.3 Student Dashboard

The student dashboard is implemented in frontend/src/pages/user/Dashboard.jsx. It presents the current profile state, CV upload/update controls, profile completeness, career identity, AI insights, and next actions. Before CV upload, it prompts the user to add a CV. After upload, it displays parsed CV availability, role inference, profile score, experience, and action buttons.

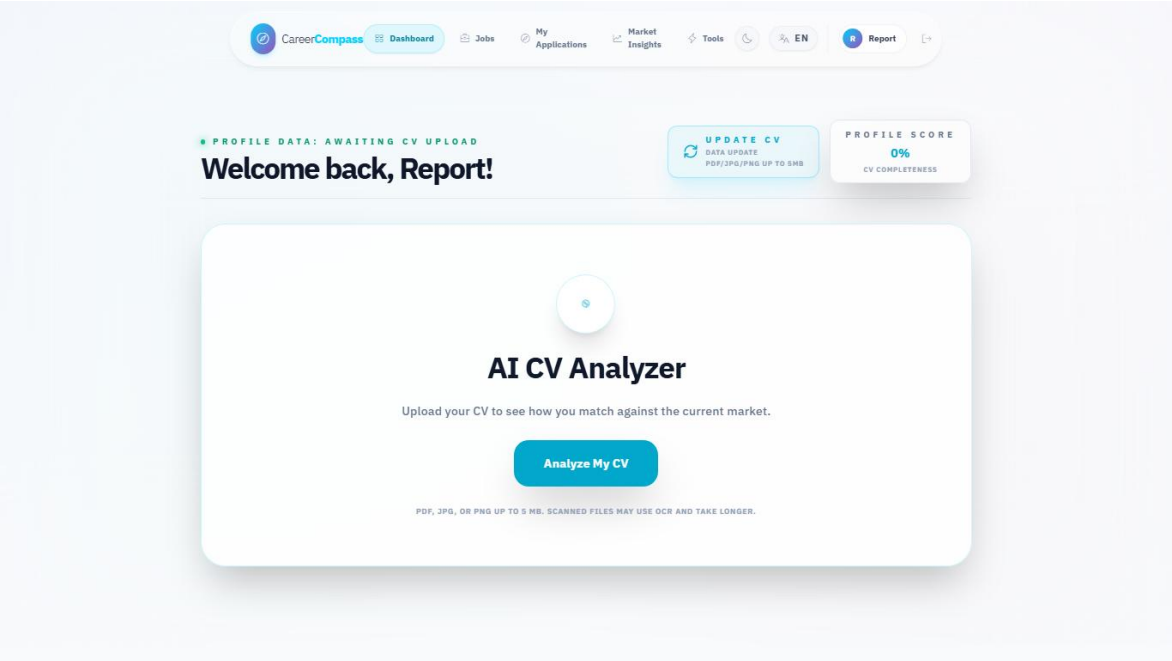


Figure 12. Student dashboard before CV upload.

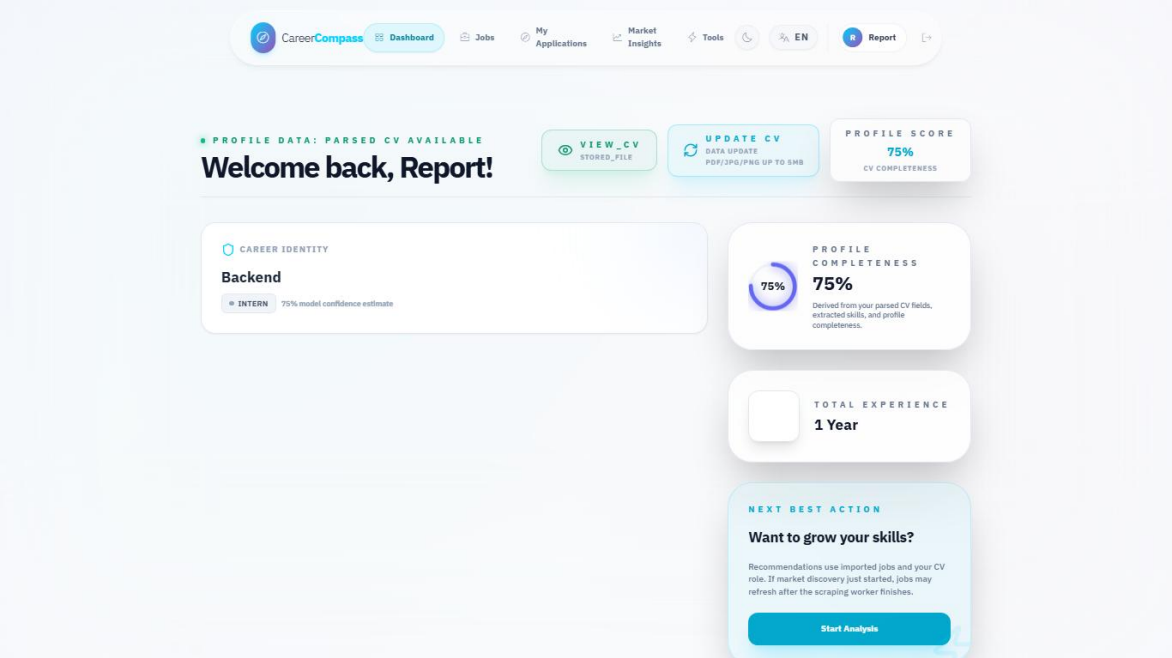


Figure 14. Dashboard after successful CV parsing.

5.4 CV Upload and Storage

CvUploadRequest requires a cv file and accepts PDF, JPEG, JPG, and PNG files up to 5 MB. The frontend appends the selected file as cv in a FormData object. CvController calls the CV processing service, persists the file path and metadata, and returns a unified user resource.

CV storage is handled as a private file workflow. The system supports signed download URLs, which is a better demo posture than public file exposure. OWASP recommends validating uploaded file type, extension, size, and storage handling carefully [27].

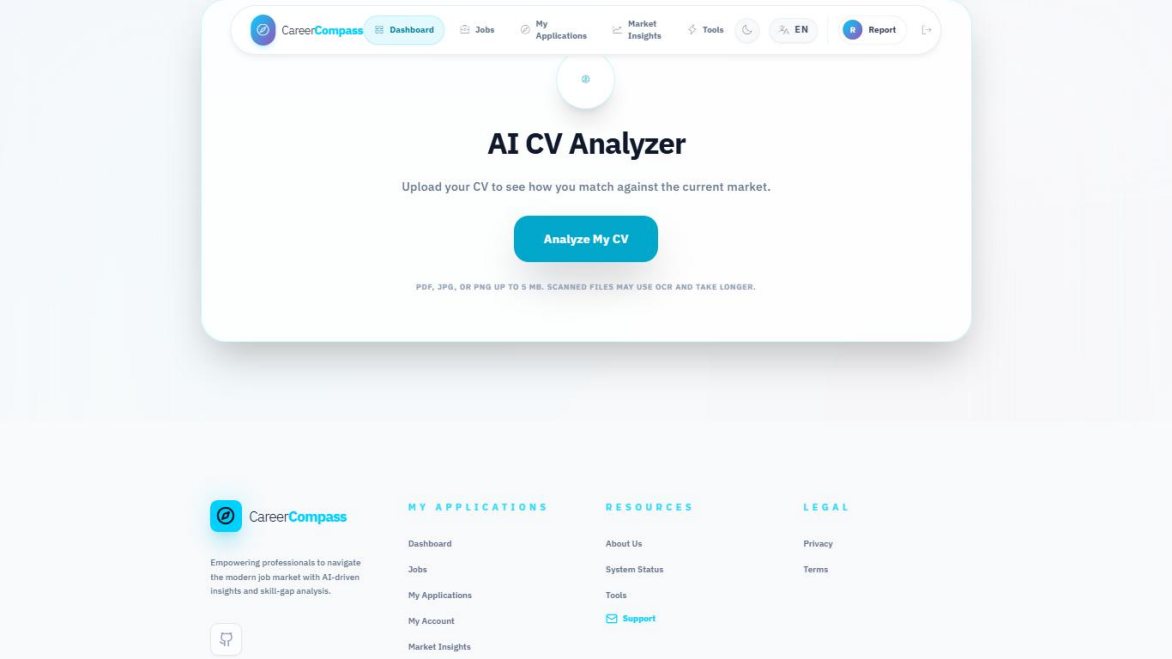


Figure 13. CV upload user interface.

5.5 CV Parsing and Skill Extraction

The CV processing flow sends the file to the AI CV Analyzer, receives parsed data, synchronizes skills, updates profile fields, and stores CV analysis metadata. The implementation handles multiple parsing statuses honestly. If analysis times out, fails, or finds no readable text, the backend returns warnings and preserves existing profile details rather than silently replacing data with low-quality output.

5.6 Profile and Skills Management

The profile page reads normalized user data, profile fields, experiences, skills, and CV analysis. The system distinguishes user fields, profile fields, extracted skills, predicted role, seniority, and completeness score. Skill synchronization is handled through backend services rather than only frontend state.

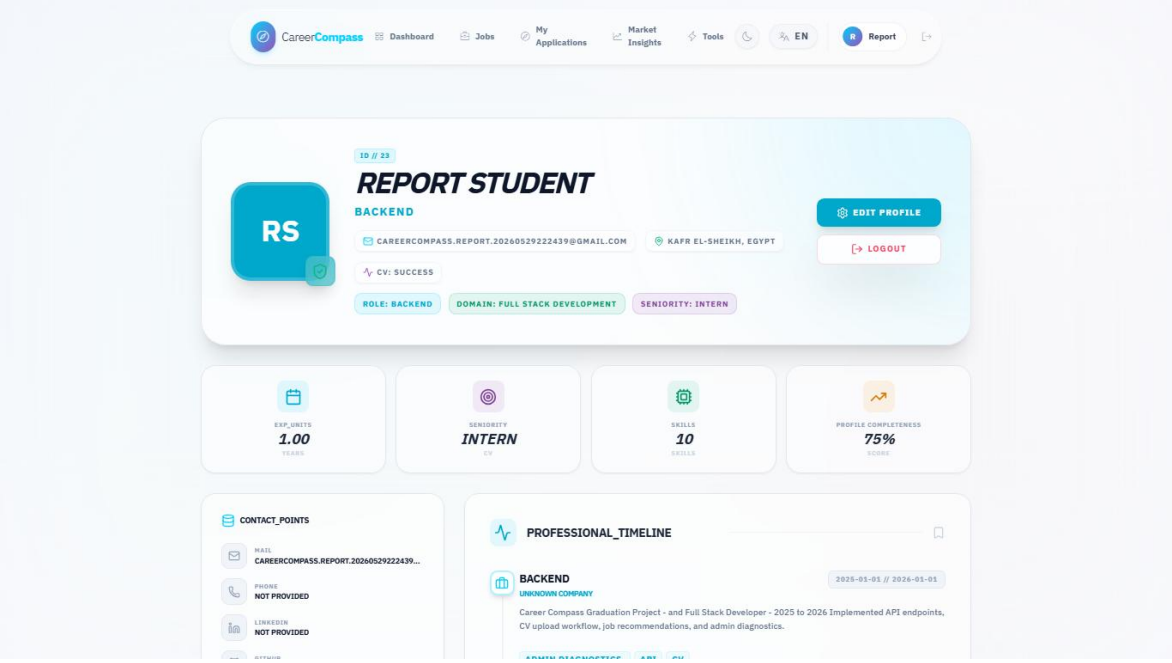


Figure 15. Extracted profile and skills page.

5.7 Job Data Model

Jobs are represented in the backend through job posting models and migrations. Fields include title, company, description/requirements, URL, source, and metadata. The seeders and import controllers enforce quality gates and uniqueness rules, including a title/company uniqueness constraint that prevented duplicate seed insertion during validation.

5.8 AI Job Miner and Scraping Sources

The job miner exposes a FastAPI service and imports jobs using configured sources. The backend protects scraper import routes with an internal service token. Admin pages expose source diagnostics, source status, testing, and target role management. The project differentiates demo/local sources, API sources, and HTML/scraping sources instead of claiming complete market coverage.

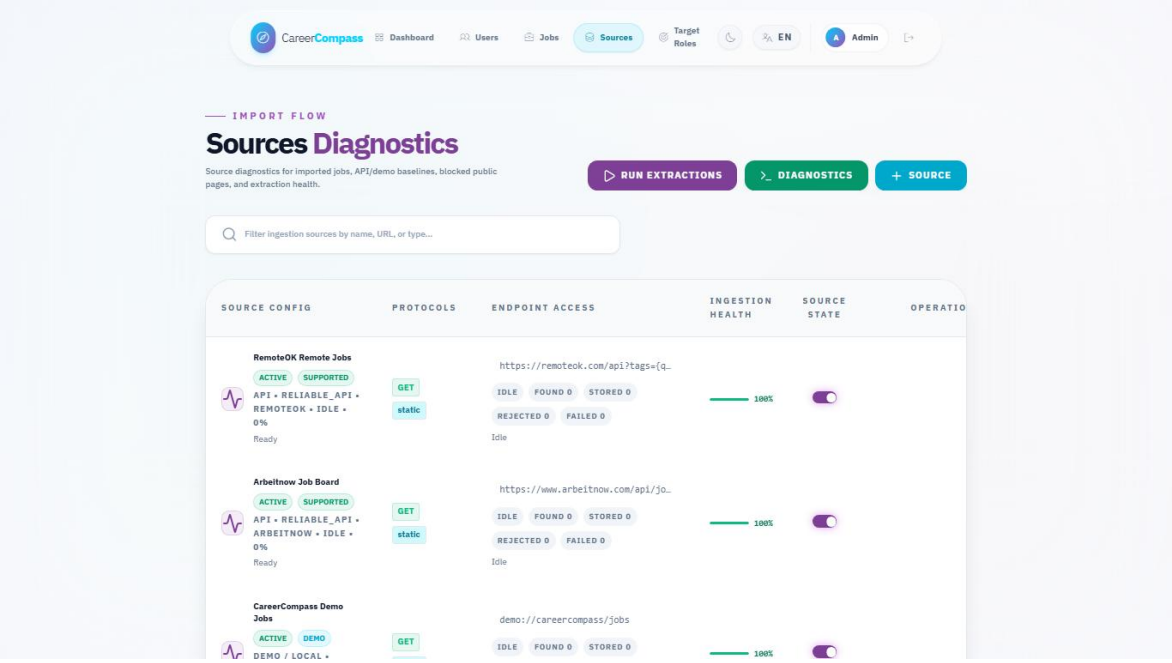


Figure 24. Admin sources diagnostics page.

5.9 Job Recommendations

The jobs page requests recommended jobs when no manual search query is active. Recommendations are based on CV/profile context when available. Matching combines normalized database data with semantic and TF-IDF-style comparison where available. TF-IDF represents text using term frequency and inverse document frequency weighting [19], while cosine similarity compares vector orientation [20].

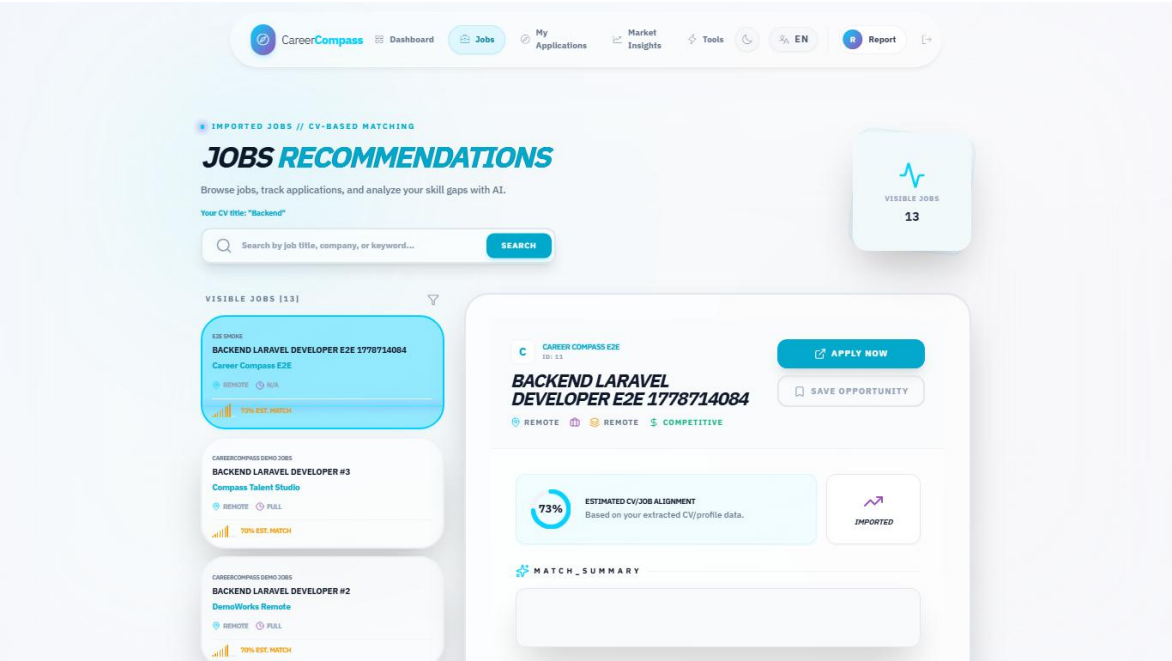


Figure 16. Jobs recommendations page.

5.10 Gap Analysis

Gap analysis compares a selected job or target role against the user's profile and extracted skills. It returns matched skills, critical/missing skills, recommendations, match percentage, and roadmap-like guidance. The frontend displays these outputs in an explainable layout rather than a single opaque score.

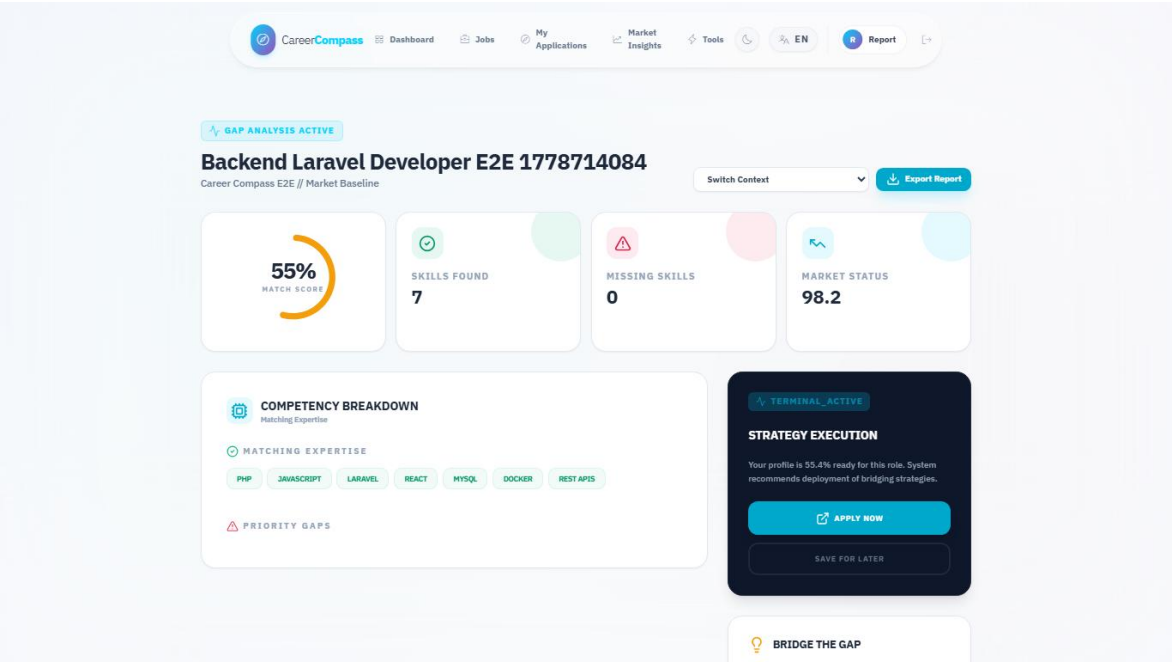


Figure 18. Gap analysis page.

5.11 Application Tracker

The application tracker is implemented through ApplicationController, ApplicationTrackerService, and frontend/src/pages/user/Applications.jsx. Students can save a job, update status, view counts, and delete tracked items. The backend validates job existence and allowed statuses.

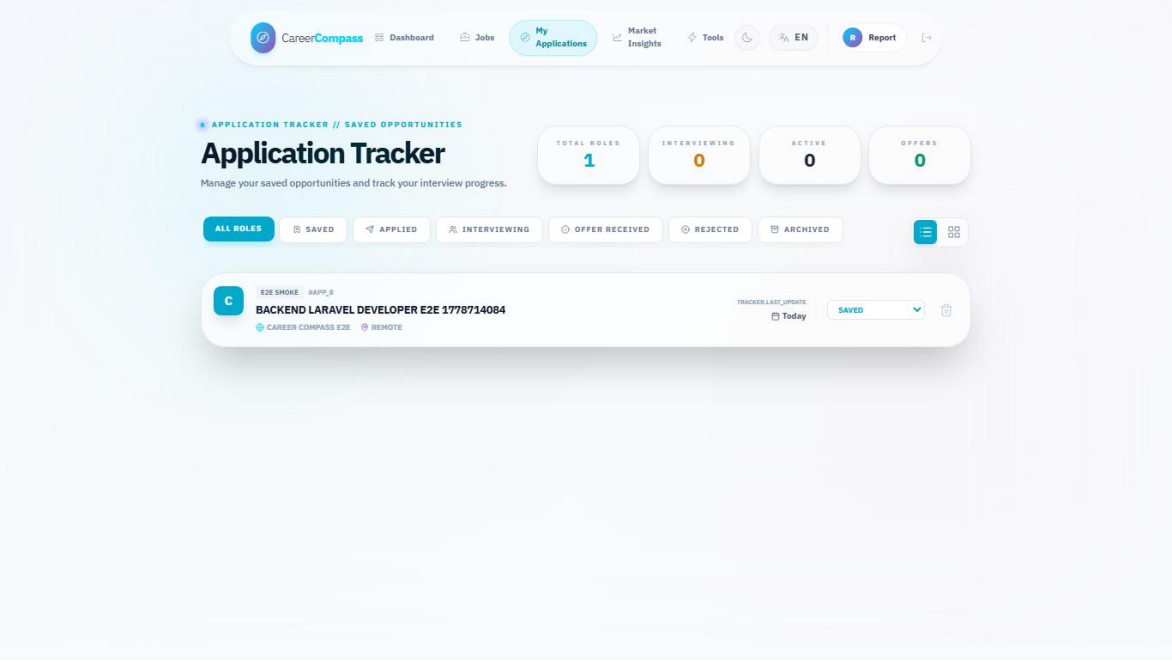


Figure 19. Applications tracker page.

5.12 Admin Dashboard

The admin dashboard summarizes users, imported jobs, active sources, target roles, health status, and scraping batch progress. It is protected by admin middleware and uses admin API routes.

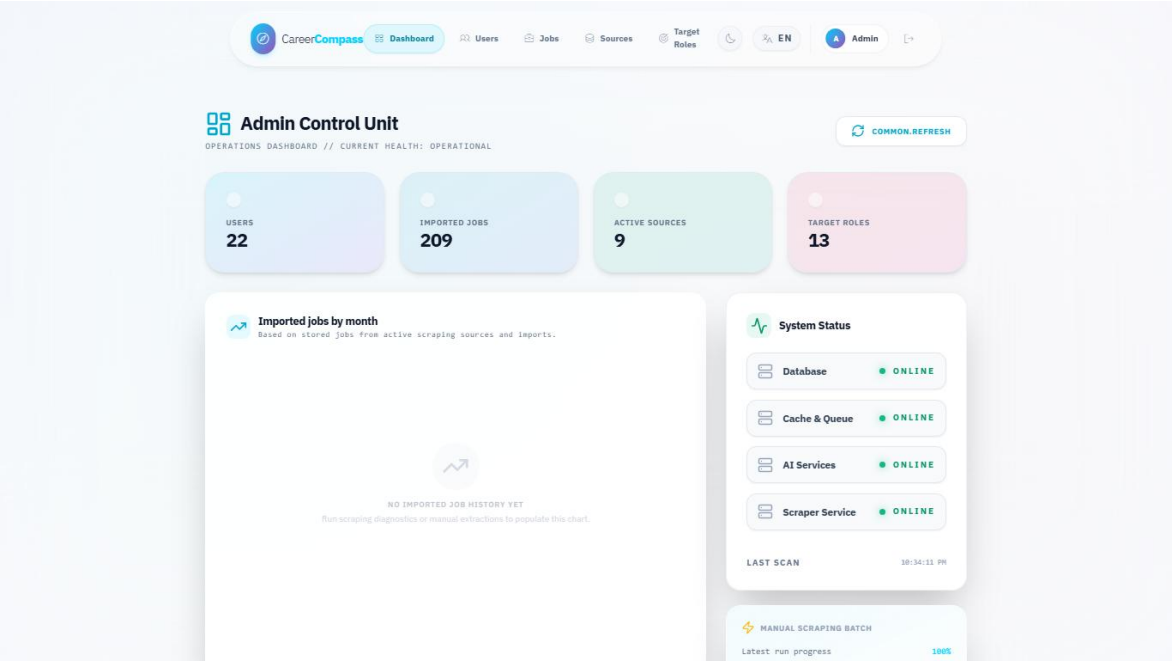


Figure 22. Admin dashboard.

5.13 Admin Source Diagnostics

The source diagnostics page lists configured scraping sources, supports source testing, and displays quality and scraping status information. The target roles page manages role names used by scraping and market discovery.

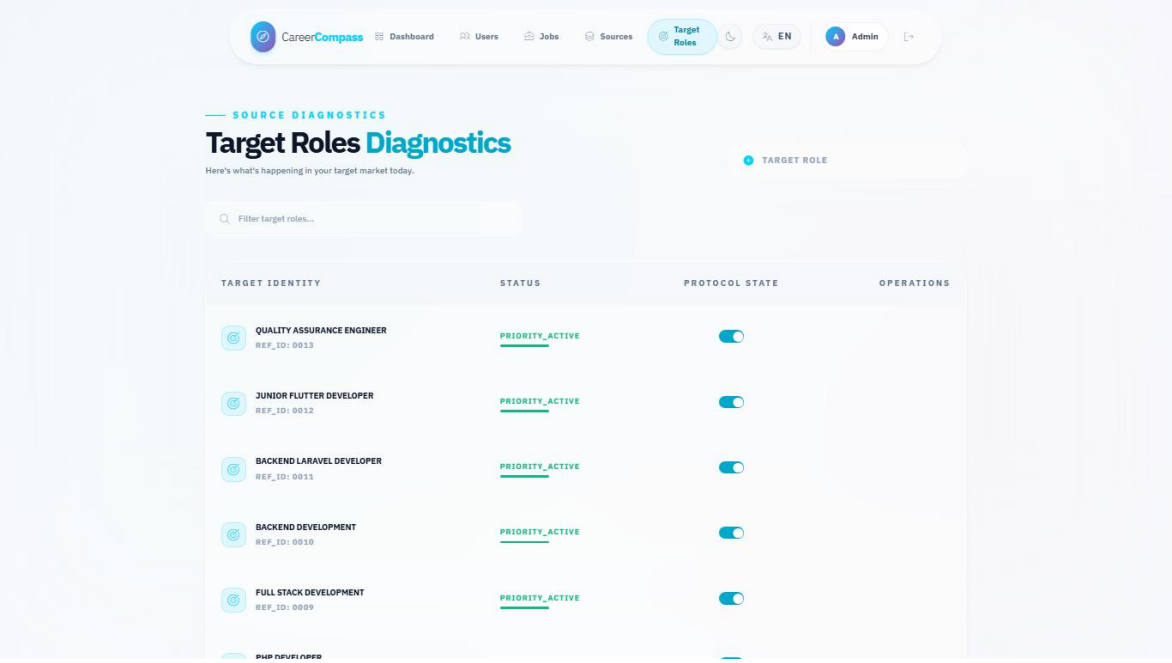


Figure 25. Admin target roles page.

5.14 System Health and Monitoring

Health endpoints include live and readiness checks. The system status page presents service state to users, while admin health data supports operational monitoring. Metrics are available for Prometheus and dashboards are available through Grafana.

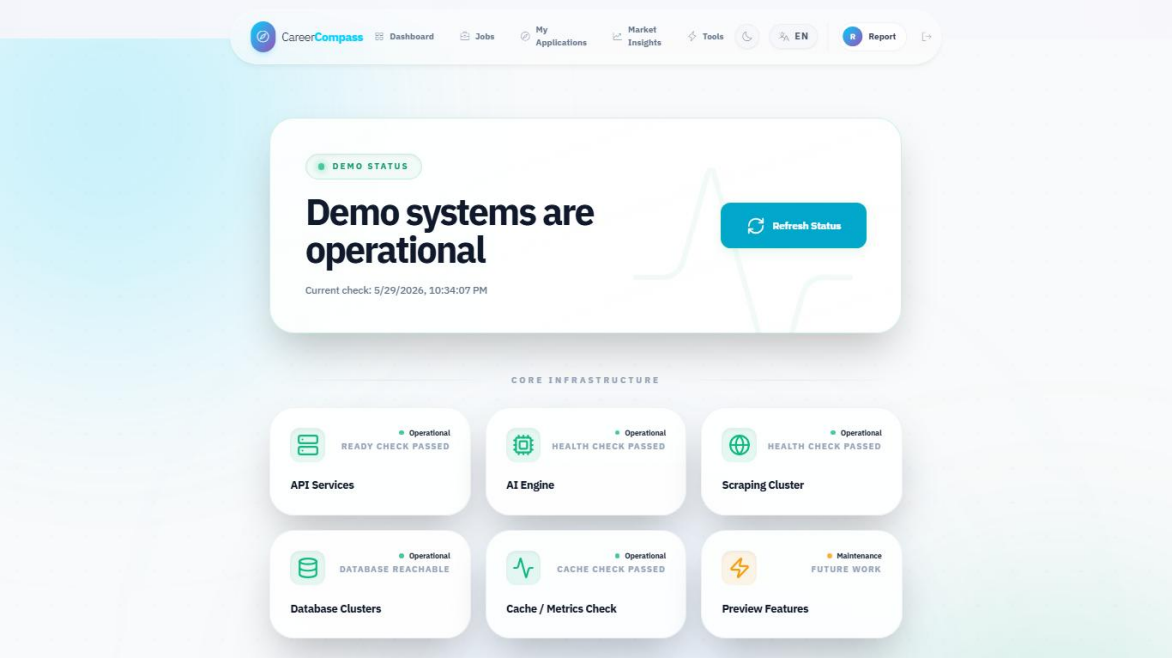


Figure 21. System status page.

5.15 Error Handling and Fallbacks

The code includes explicit handling for CV processing failures, AI gateway connection failures, validation errors, missing user data, empty job data, and unavailable services. The job recommendation and gap analysis code includes fallback behavior when AI services are not available.

5.16 Internationalization and UI Preview Modules

The frontend contains English and Arabic locale files. Preview modules include CV Builder, Mock Interview, Learning Paths, Career Planner, Mentorship, Tools Hub, and Market Intelligence. The report treats these as preview modules unless tests or implementation prove production completeness.

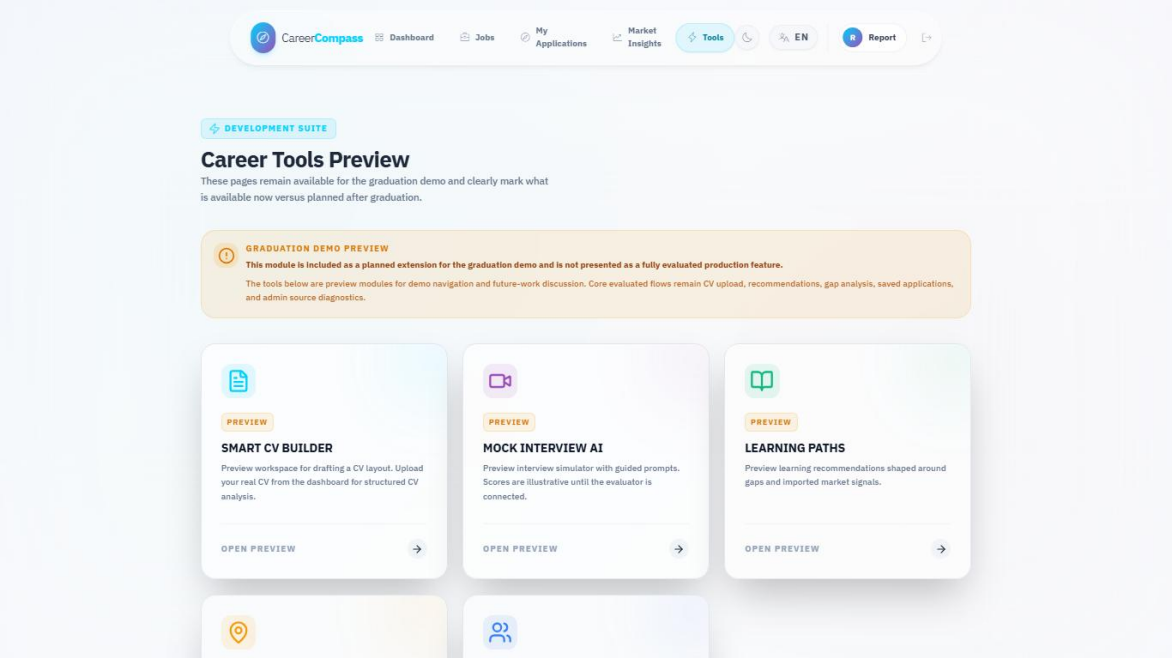
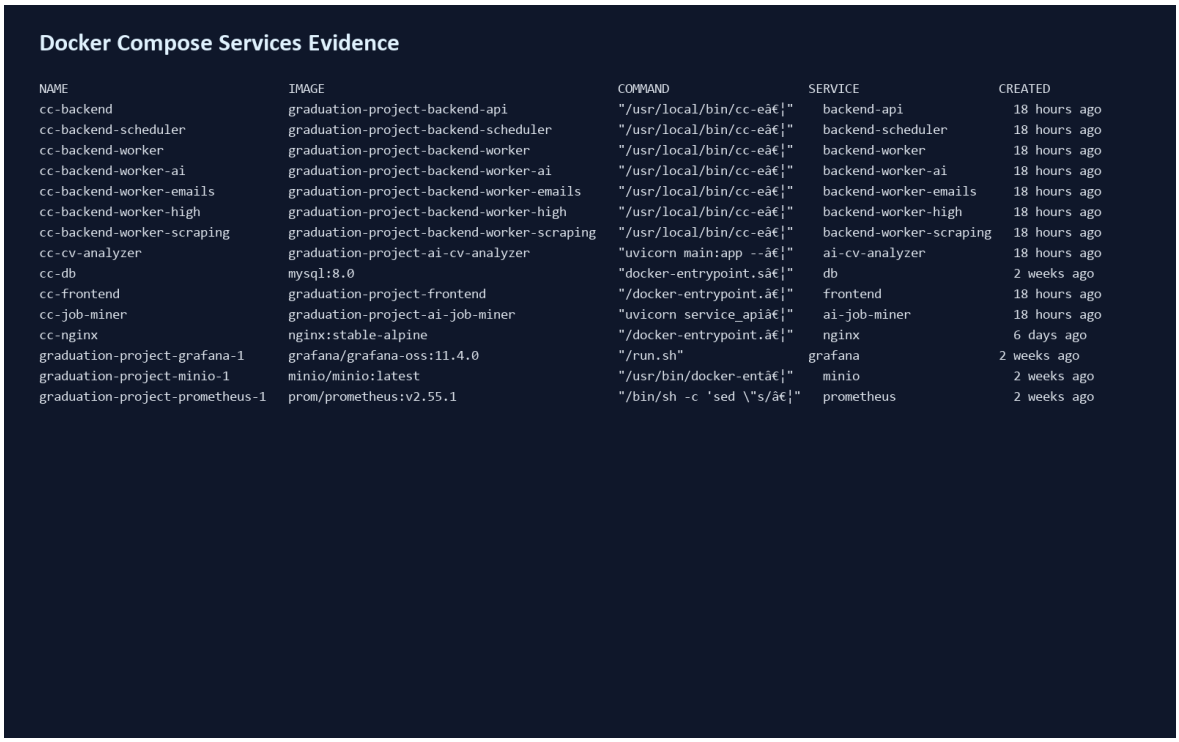


Figure 20. Tools Hub preview page.

5.17 Dockerized Runtime Flow

The runtime starts through Docker Compose. Nginx exposes the app, frontend and backend containers serve UI/API flows, backend workers process queues, Python services support AI workflows, MySQL and MinIO persist state, and monitoring services observe the stack.



Docker Compose Services Evidence

NAME	IMAGE	COMMAND	SERVICE	CREATED
cc-backend	graduation-project-backend-api	"/usr/local/bin/cc-eâ€¦"	backend-api	18 hours ago
cc-backend-scheduler	graduation-project-backend-scheduler	"/usr/local/bin/cc-eâ€¦"	backend-scheduler	18 hours ago
cc-backend-worker	graduation-project-backend-worker	"/usr/local/bin/cc-eâ€¦"	backend-worker	18 hours ago
cc-backend-worker-ai	graduation-project-backend-worker-ai	"/usr/local/bin/cc-eâ€¦"	backend-worker-ai	18 hours ago
cc-backend-worker-emails	graduation-project-backend-worker-emails	"/usr/local/bin/cc-eâ€¦"	backend-worker-emails	18 hours ago
cc-backend-worker-high	graduation-project-backend-worker-high	"/usr/local/bin/cc-eâ€¦"	backend-worker-high	18 hours ago
cc-backend-worker-scraping	graduation-project-backend-worker-scraping	"/usr/local/bin/cc-eâ€¦"	backend-worker-scraping	18 hours ago
cc-cv-analyzer	graduation-project-ai-cv-analyzer	"uvicorn main:app --â€¦"	ai-cv-analyzer	18 hours ago
cc-db	mysql:8.0	"docker-entrypoint.sâ€¦"	db	2 weeks ago
cc-frontend	graduation-project-frontend	"/docker-entrypoint.â€¦"	frontend	18 hours ago
cc-job-miner	graduation-project-ai-job-miner	"uvicorn service_apiâ€¦"	ai-job-miner	18 hours ago
cc-nginx	nginx:stable-alpine	"/docker-entrypoint.â€¦"	nginx	6 days ago
graduation-project-grafana-1	grafana/grafana-oss:11.4.0	"/run.sh"	grafana	2 weeks ago
graduation-project-minio-1	minio/minio:latest	"/usr/bin/docker-entâ€¦"	minio	2 weeks ago
graduation-project-prometheus-1	prom/prometheus:v2.55.1	"/bin/sh -c 'sed \"s/â€¦"	prometheus	2 weeks ago

Figure 26. Docker services evidence.

Chapter 6: Testing and Evaluation

6.1 Introduction

Evaluation was performed using repository-aware commands and browser evidence. The goal was to verify that each major part of the graduation/demo system runs and to document limitations honestly.

6.2 Testing Strategy

The testing strategy combined automated tests, build checks, configuration checks, service health probes, and manual functional evaluation. Automated tests provide repeatable evidence. Screenshots provide visible workflow evidence. Manual tables document behavior that is difficult to fully automate in the available environment.

6.3 Backend Testing

Backend validation was executed inside the backend container. Composer dependencies were already installed. php artisan config:clear, php artisan route:list, migrations, and tests passed. The route list confirmed 131 routes. The Laravel test suite passed with 39 tests and 297 assertions.

6.4 Frontend Testing

The frontend was validated using the existing frontend/node_modules and the bundled Node runtime. ESLint passed with 9 warnings and 0 errors. The warnings were related to React fast-refresh export conventions and hook dependency notes. The Vite production build passed and transformed 2904 modules.

6.5 Python Services Testing

The AI Job Miner pytest suite passed with 75 tests. Python syntax compilation passed for both AI services. The AI CV Analyzer pytest command could not run because pytest was not installed in that container; this is recorded as a skipped/blocked validation step rather than a failure of the application code.

6.6 Docker and Integration Testing

Docker Compose configuration validation passed for both development and production overlay configurations. A full compose build/start was attempted; the initial full build exceeded the 15-minute command timeout, but the stack continued building and was later brought up successfully with targeted frontend/Nginx rebuild/start. All main containers reached healthy or running state during final checks.

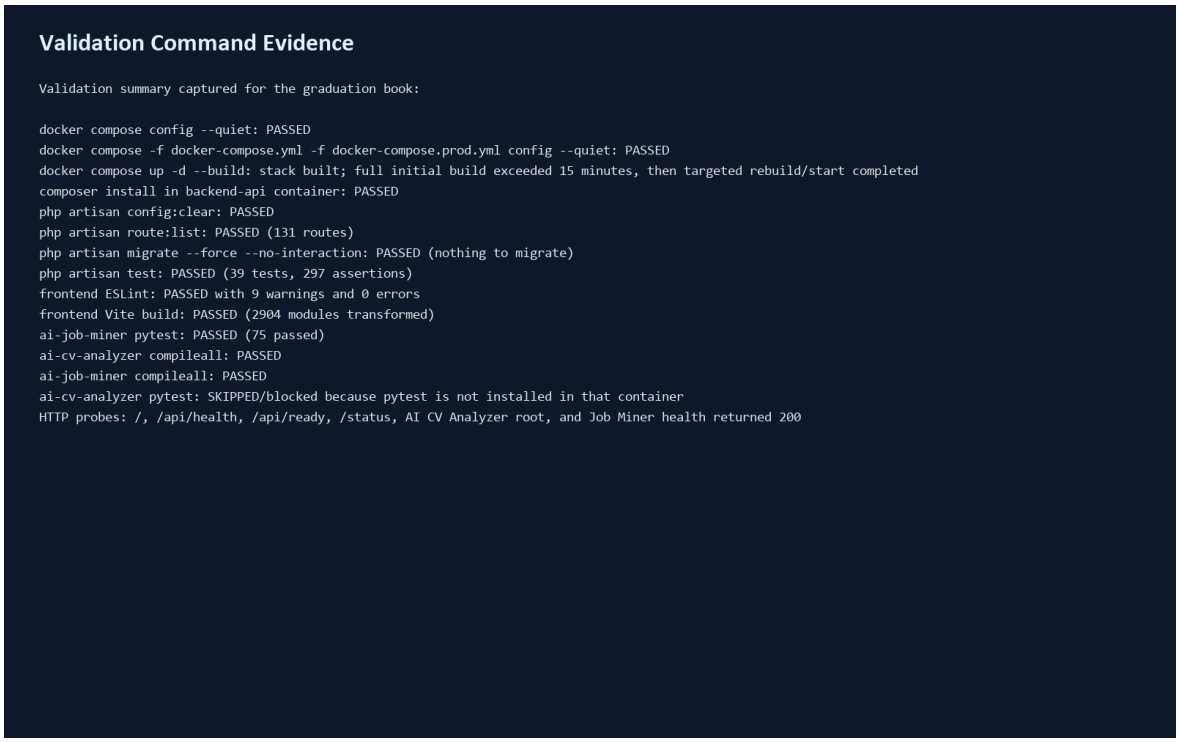


Figure 27. Validation command evidence.

6.7 CI/CD Validation

GitHub Actions workflow files were reviewed as part of repository inspection. A live GitHub Actions status screenshot was not captured before the draft PR because PR checks only become meaningful after the branch is pushed and GitHub schedules workflows. The manual review checklist asks the team to inspect CI status on the opened draft PR.

6.8 CV Analyzer Mini Dataset Evaluation

A sample PDF CV was generated for the screenshot workflow and uploaded through the running system. The upload succeeded, and the dashboard showed parsed CV data, backend role inference, extracted skills, and profile completeness. To strengthen the evaluation beyond that smoke test, this revision adds a mini synthetic dataset under docs/graduation-book/evaluation/.

The mini CV evaluation is explicitly offline and deterministic. It uses fake CV text, expected skill labels, and a keyword/role inference evaluator. It does not claim live model accuracy. The live AI CV Analyzer endpoint can be added to this mini-evaluation later, but the current document records only metrics that were actually computed from the synthetic dataset.

6.9 Recommendation Mini Dataset Evaluation

The recommendation mini evaluation ranks synthetic jobs for each synthetic CV using skill overlap plus domain and seniority bonuses. This validates the recommendation concept and provides a repeatable regression check for report evidence. It is not a production recommender benchmark, and the report does not claim complete job-market coverage.

6.10 Gap Analysis Mini Dataset Evaluation

The gap-analysis mini evaluation compares expected matched and missing skills with computed matched and missing skills for selected CV/job pairs. This directly validates the explanation structure used by the gap-analysis workflow: matched skills should be shown separately from missing skills.

Mini Dataset Files

The mini evaluation uses fake synthetic CVs and fake synthetic job records stored under docs/graduation-book/evaluation/. It is intentionally small and preliminary. It is useful for graduation validation and regression checks, but it is not statistically representative and should not be used as a production benchmark.

Sample ID	Expected Role	Seniority	Domain	Expected Skills
cv_backend_laravel	Backend Laravel Developer	junior	backend_web	PHP, Laravel, MySQL, REST API, Docker, Git
cv_frontend_react	Frontend React Developer	intern	frontend_web	React, JavaScript, Vite, HTML, CSS, API integration
cv_data_ml	Data and ML Student	student	data_ml	Python, pandas, scikit-learn, NLP, data analysis
cv_full_stack	Full Stack Developer	junior	full_stack_web	Laravel, React, MySQL, Docker, REST API, Git
cv_qa_testing	QA Testing Engineer	intern	quality_assurance	testing, test cases, pytest, API testing, bug reporting

Table 6. Mini CV dataset.

Job ID	Title	Domain	Required Skills
job_laravel_backend	Junior Laravel Backend Developer	backend_web	PHP, Laravel, MySQL, REST API, Docker, Git
job_react_frontend	React Frontend Intern	frontend_web	React, JavaScript, Vite, HTML, CSS, API integration
job_full_stack_web	Full Stack Web Developer	full_stack_web	Laravel, React, MySQL, Docker, REST API, Git
job_data_analyst	Junior Data Analyst	data_ml	Python, pandas, scikit-learn, NLP, data analysis
job_qa_intern	QA Automation Intern	quality_assurance	testing, test cases, pytest, API testing, bug reporting
job_devops_docker	DevOps Docker Intern	devops	Docker, Git, CI, Linux, monitoring
job_php_api	PHP API Developer	backend_web	PHP, Laravel, REST API, MySQL, Git, Docker
job_nlp_assistant	NLP Assistant Intern	data_ml	Python, NLP, scikit-learn, data analysis, testing

Table 7. Mini job dataset.

Metric Definitions

Skill precision measures how many extracted skills are expected labels. Skill recall measures how many expected skills were extracted. Skill F1 is the harmonic mean of precision and recall [30]. Recommendation top-1 and top-3 relevance compare ranked jobs against manual relevance labels. Gap agreement compares computed matched/missing skills against expected matched/missing skills.

Area	Metric	Value	Notes
CV offline	Macro skill precision	1.000	Keyword extraction over synthetic CV text

Area	Metric	Value	Notes
CV offline	Macro skill recall	1.000	Compared with expected skill labels
CV offline	Macro skill F1	1.000	F1 computed from precision/recall [30]
CV offline	Role match rate	1.000	Rule-based role inference on synthetic data
CV offline	Seniority match rate	1.000	Rule-based seniority inference
CV offline	Domain match rate	1.000	Rule-based domain inference
Recommendation offline	Top-1 relevance	1.000	Top recommendation belongs to manual relevant set
Recommendation offline	Top-3 relevance	1.000	Any top-3 job belongs to manual relevant set
Recommendation offline	Mean precision@3	0.800	Relevant jobs among top three
Gap offline	Matched skill agreement F1	1.000	Computed matched skills vs. expected matched skills
Gap offline	Missing skill agreement F1	1.000	Computed missing skills vs. expected missing skills

Table 8. Mini evaluation metrics.

Recommendation Ranking Details

CV Sample	Expected Relevant Jobs	Top 3 Offline Recommendations	P@3
cv_backend_laravel	job_laravel_backend, job_php_api, job_full_stack_web, job_devops_docker	job_laravel_backend, job_php_api, job_full_stack_web	1.000
cv_frontend_react	job_react_frontend, job_full_stack_web	job_react_frontend, job_full_stack_web, job_qa_intern	0.667
cv_data_ml	job_data_analyst, job_nlp_assistant	job_data_analyst, job_nlp_assistant, job_laravel_backend	0.667
cv_full_stack	job_full_stack_web, job_laravel_backend, job_php_api, job_react_frontend, job_devops_docker	job_full_stack_web, job_laravel_backend, job_php_api	1.000
cv_qa_testing	job_qa_intern, job_nlp_assistant	job_qa_intern, job_nlp_assistant, job_react_frontend	0.667

Table 9. Recommendation ranking details.

Gap Analysis Pair Details

CV / Job Pair	Matched Skills	Missing Skills	Agreement
cv_backend_laravel -> job_laravel_backend	Docker, Git, Laravel, MySQL, PHP, REST API	None	matched F1=1.000; missing F1=1.000
cv_frontend_react -> job_full_stack_web	React	Docker, Git, Laravel, MySQL, REST API	matched F1=1.000; missing F1=1.000

CV / Job Pair	Matched Skills	Missing Skills	Agreement
cv_data_ml -> job_nlp_assistant	NLP, Python, data analysis, scikit-learn	testing	matched F1=1.000; missing F1=1.000
cv_qa_testing -> job_qa_intern	API testing, bug reporting, pytest, test cases, testing	None	matched F1=1.000; missing F1=1.000
cv_full_stack -> job_react_frontend	React	API integration, CSS, HTML, JavaScript, Vite	matched F1=1.000; missing F1=1.000

Table 10. Gap analysis pair details.

6.11 Job Miner Evaluation

The AI Job Miner test suite passed with 75 tests. Admin source diagnostics displayed active sources and source state. The job database contained imported/demo jobs visible in the admin dashboard. Because external job sources can change or throttle scraping, long-term data quality evaluation should be repeated near the final defense.

6.12 Application Tracker Evaluation

The application tracker was evaluated by saving a selected job to the tracker and loading the Applications page. The screenshot shows saved opportunity state. Backend tests also include application tracker behavior.

6.13 Admin Dashboard Evaluation

Admin login and admin dashboard access were tested with the demo admin account. The admin dashboard, admin jobs, admin sources, and admin target roles pages were captured. The dashboard displayed 22 users, 209 imported jobs, 9 active sources, and 13 target roles at capture time.

6.14 Performance Observations

The local Docker stack is heavy because it runs frontend, backend, multiple Laravel workers, MySQL, MinIO, two Python AI services, Prometheus, and Grafana. Initial full build can exceed a short command timeout on a Windows laptop. Once images are built, targeted service startup and HTTP checks are practical for a graduation demo.

6.15 Evaluation Limitations

- The AI CV Analyzer pytest suite was not executed because pytest was absent in that container.
- The browser CV upload remains a smoke test, and the mini dataset is synthetic rather than statistically representative.
- The recommendation score shown in screenshots is an estimated local demo output.
- External scraping reliability depends on source availability and changing website/API behavior.
- Production security, privacy, and performance audits remain future work.

6.16 Summary of Results

Area	Command or Scenario	Result	Evidence
Docker config	docker compose config --quiet	Passed	Terminal evidence
Docker prod config	docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet	Passed	Terminal evidence

Area	Command or Scenario	Result	Evidence
Backend dependencies	composer install --no-interaction --prefer-dist	Passed	Command output
Backend routes	php artisan route:list	Passed, 131 routes	Command output
Backend tests	php artisan test	Passed, 39 tests, 297 assertions	Command output
Frontend lint	ESLint	Passed, 9 warnings, 0 errors	Command output
Frontend build	Vite build	Passed, 2904 modules transformed	Command output
AI Job Miner tests	python -m pytest	Passed, 75 tests	Command output
AI CV Analyzer syntax	python -m compileall .	Passed	Command output
AI CV Analyzer pytest	python -m pytest	Skipped, pytest missing	Command output
HTTP probes	/, /api/health, /api/ready, /status, AI services	200 responses	Command output

Table 11. Automated validation results.

Test ID	Module	Scenario	Status	Evidence
M-01	Authentication	Register demo user	Passed	Register screenshot/API output
M-02	Authentication	Login student	Passed	Figure 12
M-03	CV upload	Upload valid PDF	Passed	Figures 13-15
M-04	CV upload	Invalid file handling	Not Run Manual	Backend validation tests
M-05	Recommendations	Open jobs page after CV	Passed	Figure 16
M-06	Gap analysis	Analyze selected job	Passed	Figure 18
M-07	Tracker	Save job	Passed	Figure 19
M-08	Admin	Login admin and open dashboard	Passed	Figure 22
M-09	Admin sources	Open diagnostics	Passed	Figure 24
M-10	Status	Open system status page	Passed	Figure 21

Table 12. Manual functional evaluation matrix.

Test ID	Expected vs Actual Observation
M-01	Expected user creation and token return; actual user was created with an accepted Gmail-style address.
M-02	Expected dashboard access after login; actual dashboard loaded.
M-03	Expected CV acceptance and parsing; actual upload parsed successfully.
M-04	Expected validation error for invalid file; actual manual browser repetition was not run because backend validation/tests already cover the rule.
M-05	Expected recommendations with estimated matches; actual jobs page loaded.
M-06	Expected match and skill breakdown; actual gap page loaded.

Test ID	Expected vs Actual Observation
M-07	Expected saved job in tracker; actual application page loaded with saved item.
M-08	Expected admin-only dashboard; actual dashboard visible after admin login.
M-09	Expected source diagnostics; actual diagnostics page visible.
M-10	Expected health UI; actual system status page visible.

Table 13. Manual functional observations.

Chapter 7: Security and Privacy

7.1 Introduction

Security in CareerCompass is implemented for a graduation/demo context. The system includes meaningful controls, but it should not be presented as production-grade without future hardening, legal review, and operational security work.

7.2 Authentication and Authorization

User authentication uses token-based API access through Laravel. Laravel Sanctum supports API token and SPA authentication use cases [2]. CareerCompass stores an auth token in the browser and attaches it to API requests. Authorization is role-aware: student routes require authentication, while admin routes require the admin role. Authentication security should follow established password and session guidance in production [28].

7.3 Admin Access Control

Admin access is enforced server-side by admin middleware. Frontend route protection improves user experience, but the backend check is the important control. The demo admin account is generated by a seeder and should be changed or disabled in any non-demo environment.

7.4 CV File Privacy

CV files contain personal data. CareerCompass validates file type and size, stores files privately, and avoids public direct file exposure. OWASP's file upload guidance emphasizes validation, restricted storage, and safe handling [27].

7.5 Private Storage and Signed Downloads

Uploaded CV files are stored through private storage and accessed through signed or temporary URLs. MinIO/S3-compatible storage supports object-based file storage and access control patterns [9]. The graduation demo should still avoid uploading real sensitive CVs unless the environment is controlled.

7.6 Internal Service Tokens

Scraper import routes use a service token middleware. This reduces accidental public ingestion, but service tokens must be rotated, stored securely, and monitored in production.

7.7 Payload and File Validation

Laravel form requests validate registration, login, CV upload, applications, and profile updates. File validation includes MIME type, extension, and size checks. Validation reduces risk but does not replace malware scanning, deep file inspection, or content disarm in production.

7.8 Logging and Request IDs

The API client attaches request IDs, and backend logging records important events such as CV processing status and AI gateway errors. For production, logs should avoid sensitive CV content and should be retained according to a privacy policy.

7.9 Demo Security Limitations

Area	Current Demo Control	Production Hardening Needed
Admin account	Demo seeder account	Secret rotation, SSO/MFA, audit logs

Area	Current Demo Control	Production Hardening Needed
CV files	Private storage and signed URLs	Malware scanning, retention policy, consent model
Tokens	Bearer tokens	Token rotation, secure cookie strategy, revocation review
Scraper service	Internal token	Secret manager, network isolation, rate limits
Monitoring	Local Prometheus/Grafana	Auth, TLS, dashboard access control
Privacy	Local demo posture	Legal review, privacy notice, data minimization

Table 14. Security and privacy controls.

7.10 Future Production Hardening

Future work should include HTTPS-only deployment, secure cookie/session strategy, administrator MFA, centralized secrets management, object scanning, retention policies, audit logging, rate-limit review, CSRF/CORS review, dependency vulnerability scanning, and a privacy impact assessment.

Chapter 8: Conclusion and Future Work

8.1 Conclusion

CareerCompass demonstrates a practical AI-assisted career guidance workflow for a graduation project. It integrates CV parsing, profile normalization, skill extraction, imported job records, estimated recommendation, gap analysis, application tracking, admin diagnostics, and containerized deployment.

8.2 Project Achievements

- Built a working multi-service web application with frontend, backend, AI services, database, object storage, proxy, and monitoring.
- Implemented student authentication, dashboard, CV upload, profile view, jobs, gap analysis, and application tracking.
- Implemented admin dashboard, job management, source diagnostics, and target role management.
- Added tests and validation commands across backend, frontend, Python services, Docker, and HTTP probes.
- Captured browser screenshots from the running local system.
- Generated a formal report with diagrams, references, and evaluation notes.

8.3 Educational Value

The project demonstrates practical learning in software architecture, service decomposition, secure file handling, API design, database schema design, frontend state management, AI service integration, testing, Docker operations, monitoring, and academic documentation.

8.4 Current Limitations

- The system is a graduation/demo platform and not a production product.
- Recommendation and gap analysis outputs are estimates.
- AI evaluation needs larger labeled datasets and repeatable scoring.
- External scraping sources can be unstable.
- The AI CV Analyzer container needs pytest installed to run its test suite.
- Security and privacy controls need production hardening before real deployment.

8.5 Future Work

- Add a larger CV/job evaluation dataset with manual labels.
- Improve role taxonomy and skill normalization.
- Add production-grade authentication and administrator controls.
- Add malware scanning and retention policies for uploaded CV files.
- Improve recommendation explanations and calibration.
- Add automated browser end-to-end tests.
- Extend CI to run all containerized test suites consistently.
- Improve observability dashboards and alerting.
- Add deployment documentation for a secure cloud environment.

8.6 Final Remarks

CareerCompass is an original graduation project that connects academic software engineering requirements with a practical career guidance problem. Its strongest value is the integration of many

realistic system parts into one demonstrable workflow, while preserving honest language about limitations and future production work.

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Appendices

Appendix A: API Endpoint Summary

Group	Example Endpoints	Purpose
Health	/api/health, /api/ready, /api/metrics	Liveness, readiness, and Prometheus metrics.
Auth	/api/v1/register, /api/v1/login, /api/v1/logout, /api/v1/user	User identity and token lifecycle.
CV	/api/v1/upload-cv, /api/v1/user/skills, /api/v1/user/cv-analysis	CV upload, parsed analysis, and extracted skills.
Jobs	/api/v1/jobs, /api/v1/jobs/recommended, /api/v1/jobs/{id}	Job listing, details, and recommendations.
Gap Analysis	/api/v1/gap-analysis/job/{jobId}, /api/v1/gap-analysis/role/{roleId}	Skill comparison and recommendations.
Applications	/api/v1/applications	Save and update tracked opportunities.
Admin	/api/v1/admin/dashboard/stats, /api/v1/admin/jobs, /api/v1/admin/scraping-sources, /api/v1/admin/target-roles	Admin dashboards and diagnostics.
Internal Scraper	/api/jobs/import, /api/jobs/import/check, /api/proxies/active	Service-token protected import routes.

Table 15. API endpoint summary.

Appendix B: Database Tables Summary

Table	Purpose
users	Authentication identity, role, and banned status.
profiles	Student profile details and contact metadata.
skills	Canonical skill catalog.
user_skill	User-to-skill relationship and proficiency metadata.
experiences	Extracted or entered experience records.
cv_analyses	CV parsing status, predicted role, confidence, file metadata, strengths, gaps, and warnings.
job_postings	Imported/demo job data.
applications	Saved opportunities and status tracking.
scraping_sources	Admin-configured job source definitions.
target_job_roles	Target role list for scraping and market exploration.
scraping_jobs	Scraping batch execution state.

Table 16. Database tables summary.

Appendix C: Docker Services Summary

Service	Role
nginx	Public gateway and reverse proxy.
frontend	React/Vite web UI container.
backend-api	Laravel API runtime.
backend-worker variants	Queue processing for default, high, AI, emails, and scraping work.
backend-scheduler	Scheduled Laravel commands.
db	MySQL database.
minio	S3-compatible CV object storage.
ai-cv-analyzer	FastAPI CV parsing service.
ai-job-miner	FastAPI job mining service.
prometheus	Metrics collection.
grafana	Metrics visualization.

Table 17. Docker services summary.

Appendix D: Screenshots

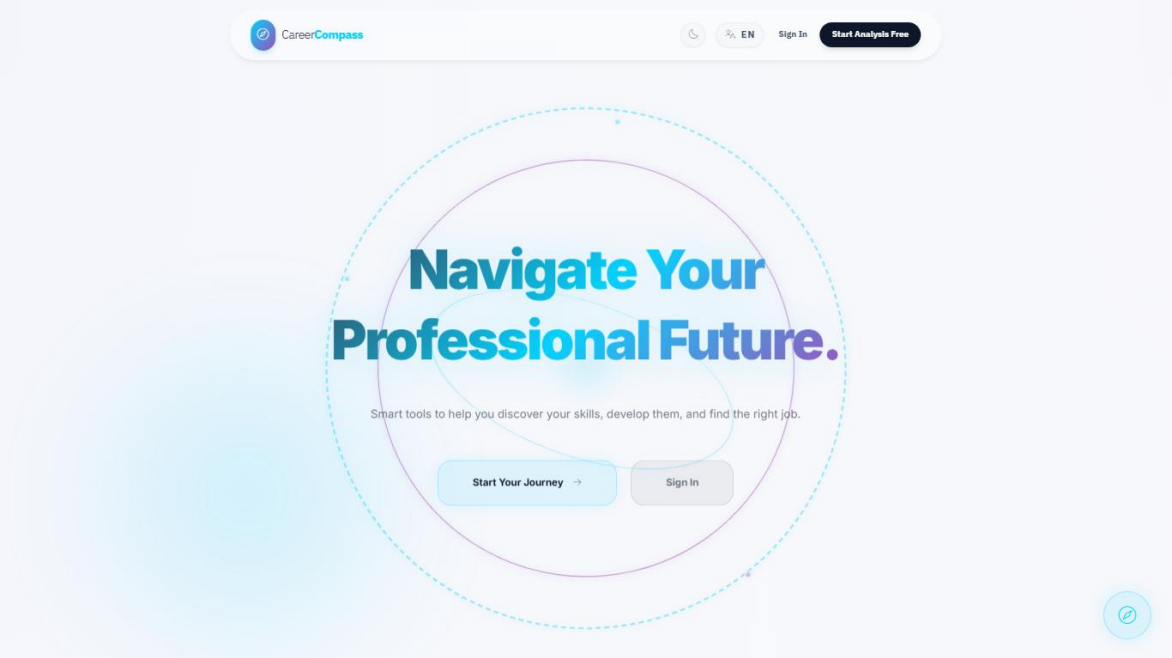

Figure 9. Home page.

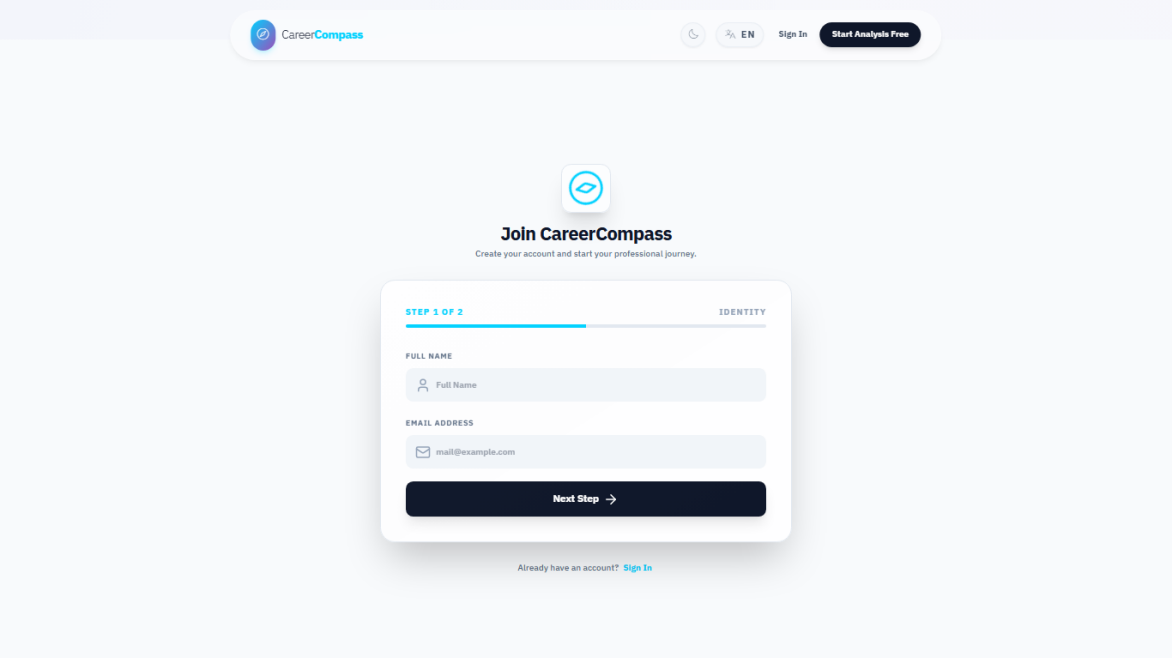

Figure 10. Register page.

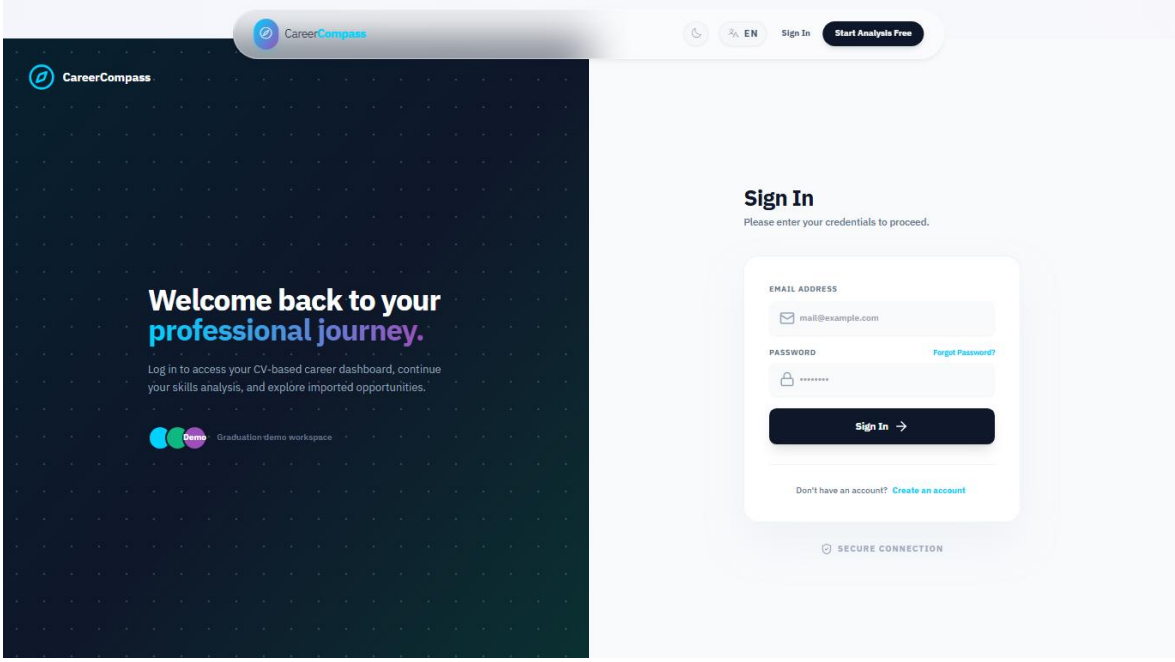

Figure 11. Login page.

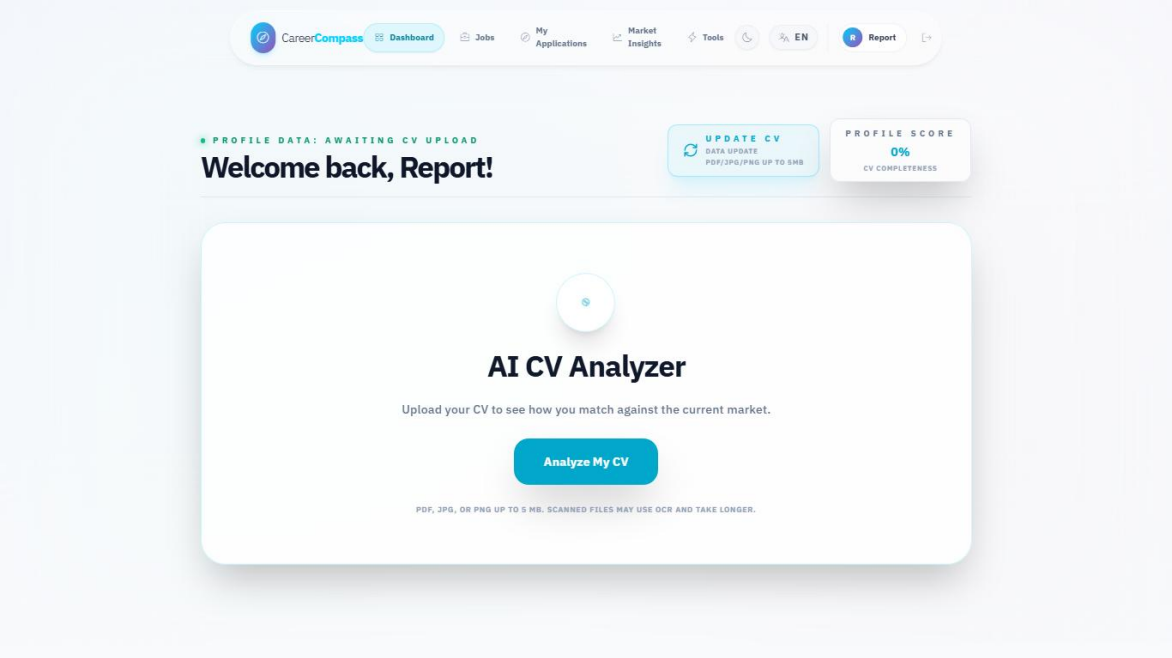

Figure 12. Student dashboard before CV upload.

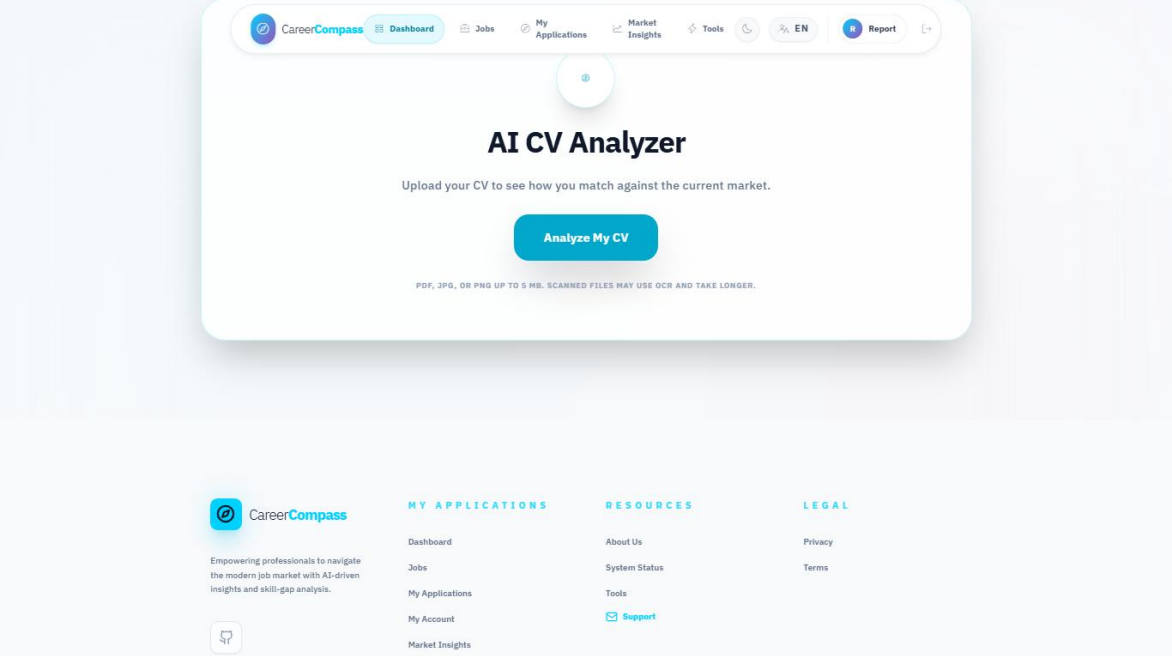

Figure 13. CV upload user interface.

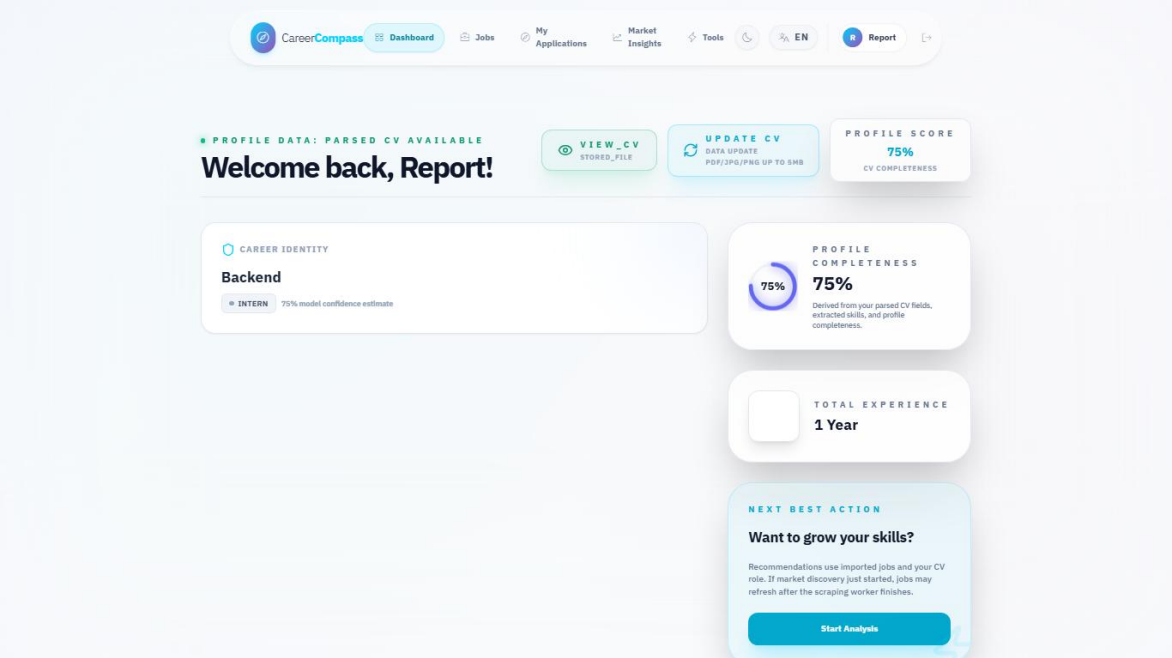

Figure 14. Dashboard after successful CV parsing.

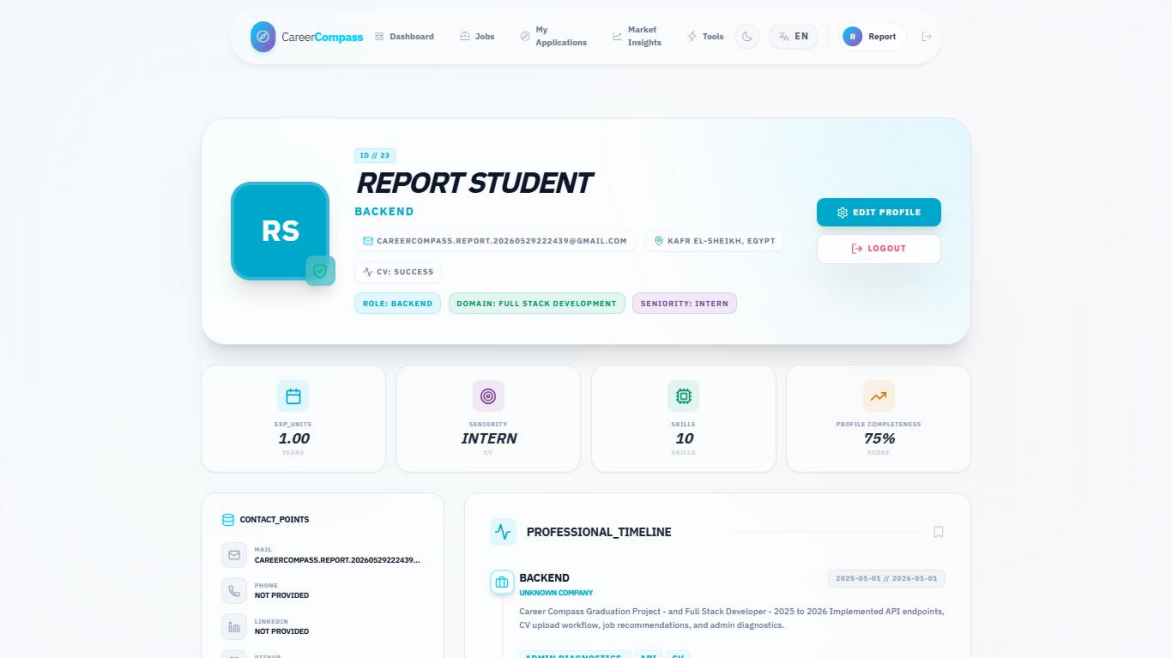

Figure 15. Extracted profile and skills page.

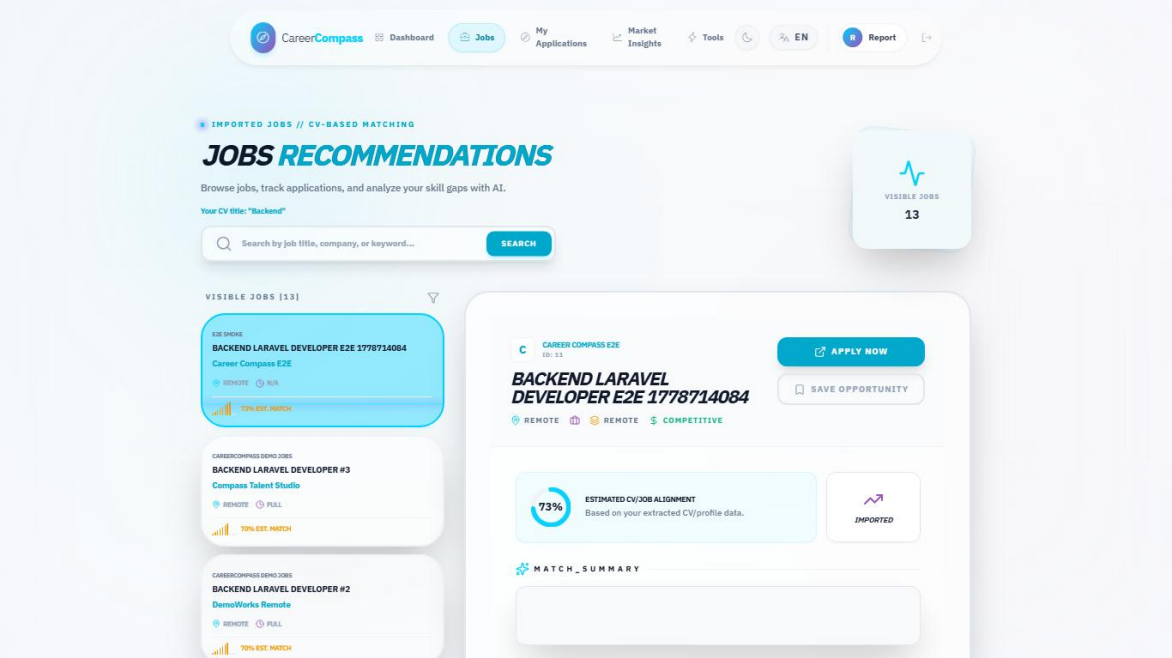

Figure 16. Jobs recommendations page.

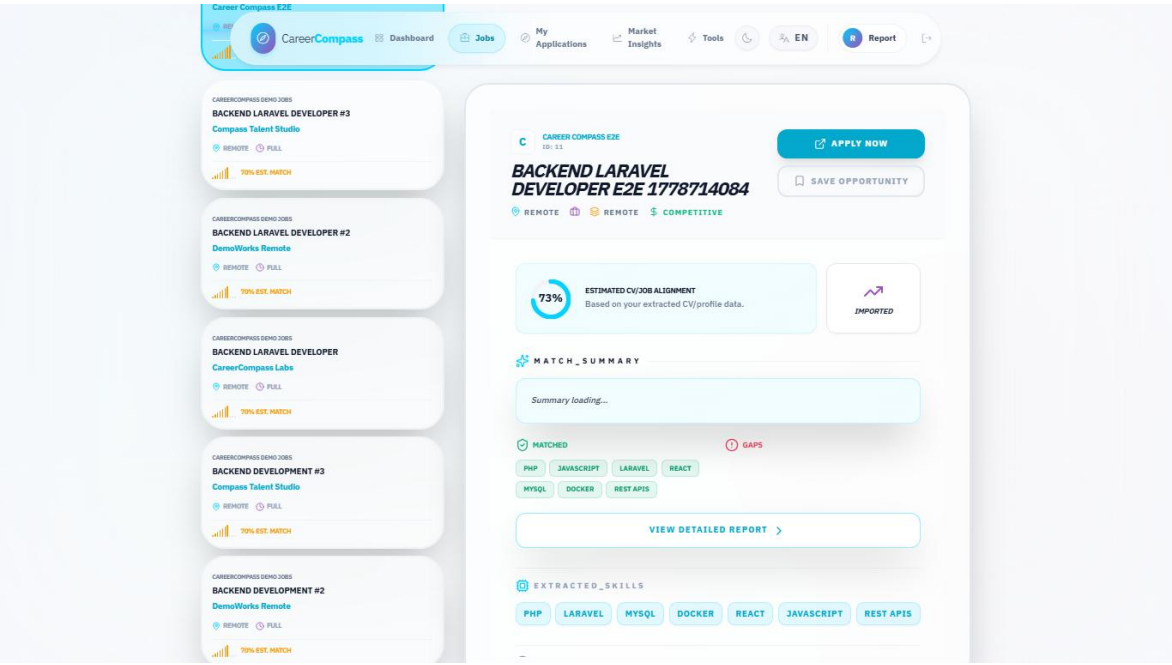

Figure 17. Job detail and inline gap panel.

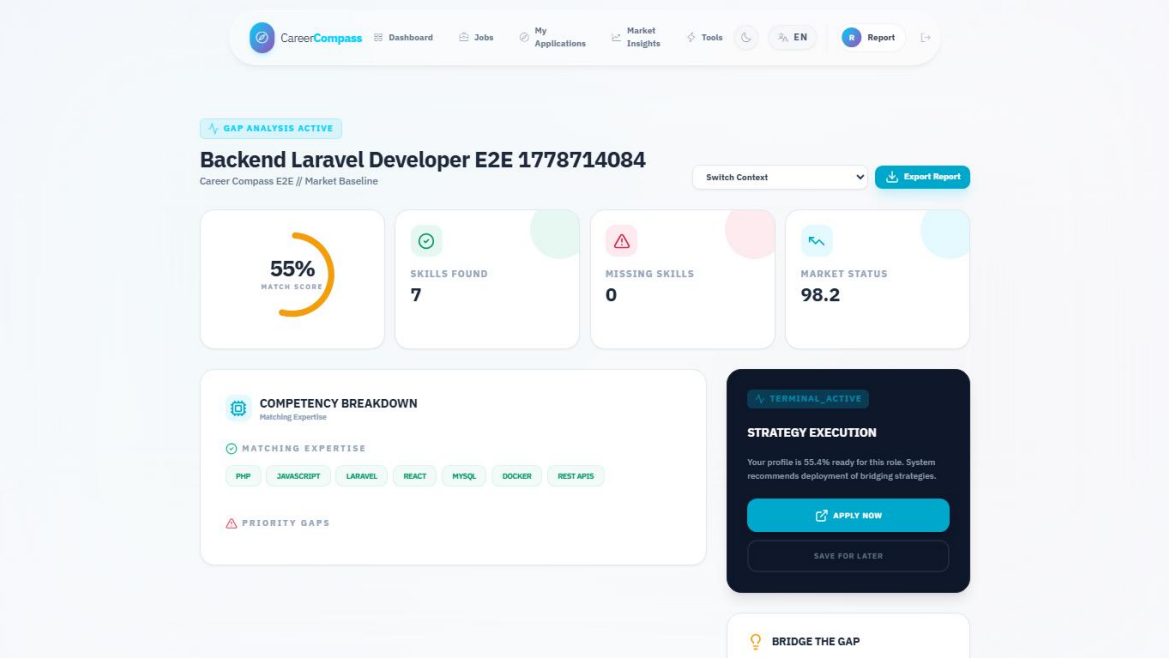

Figure 18. Gap analysis page.

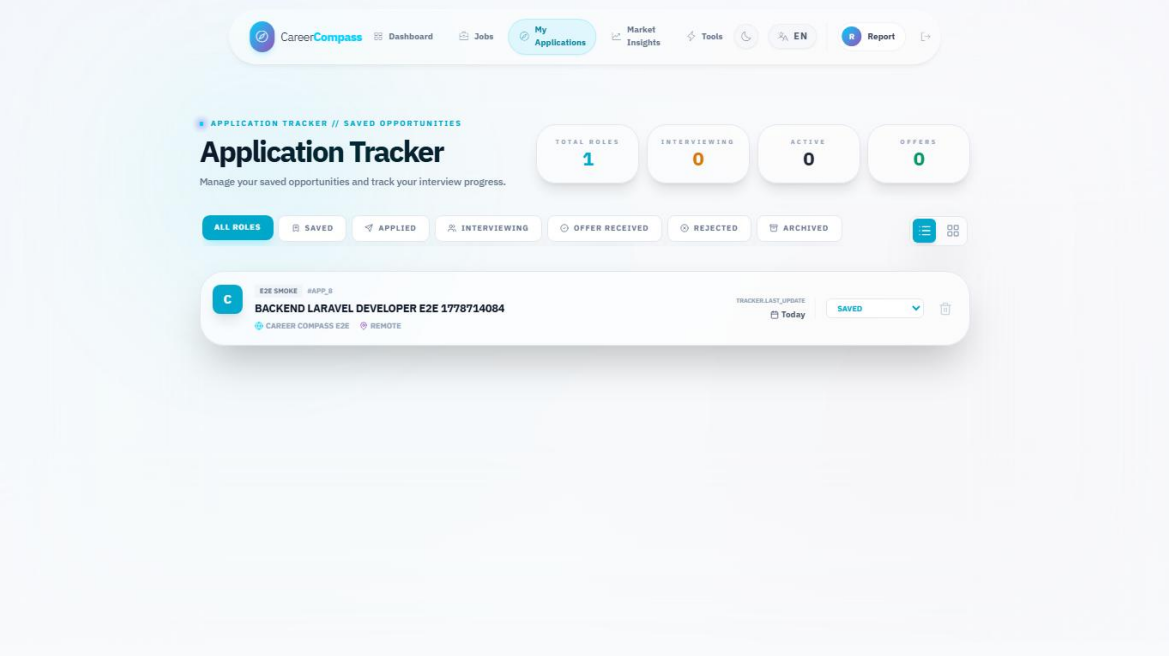

Figure 19. Applications tracker page.

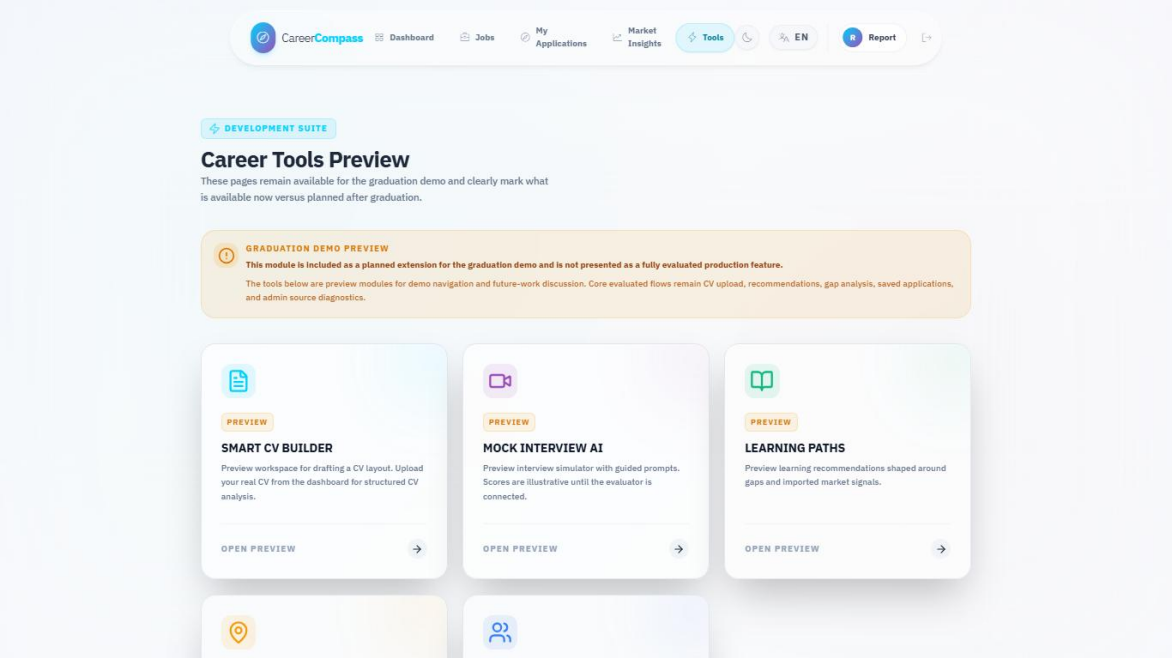

Figure 20. Tools Hub preview page.

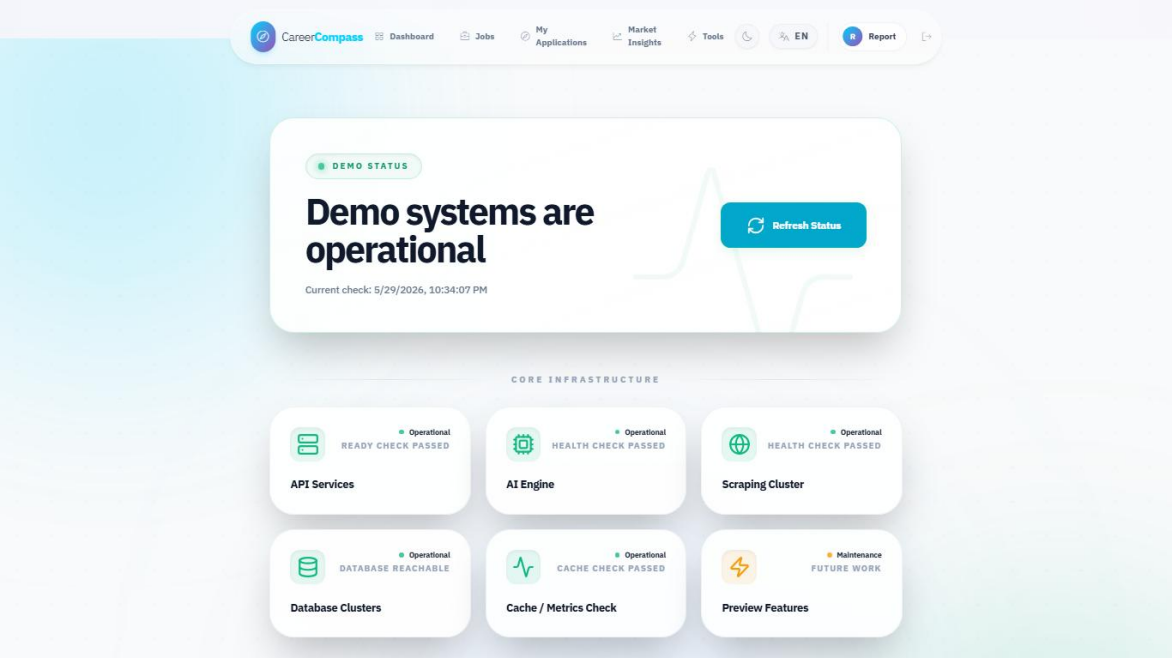

Figure 21. System status page.

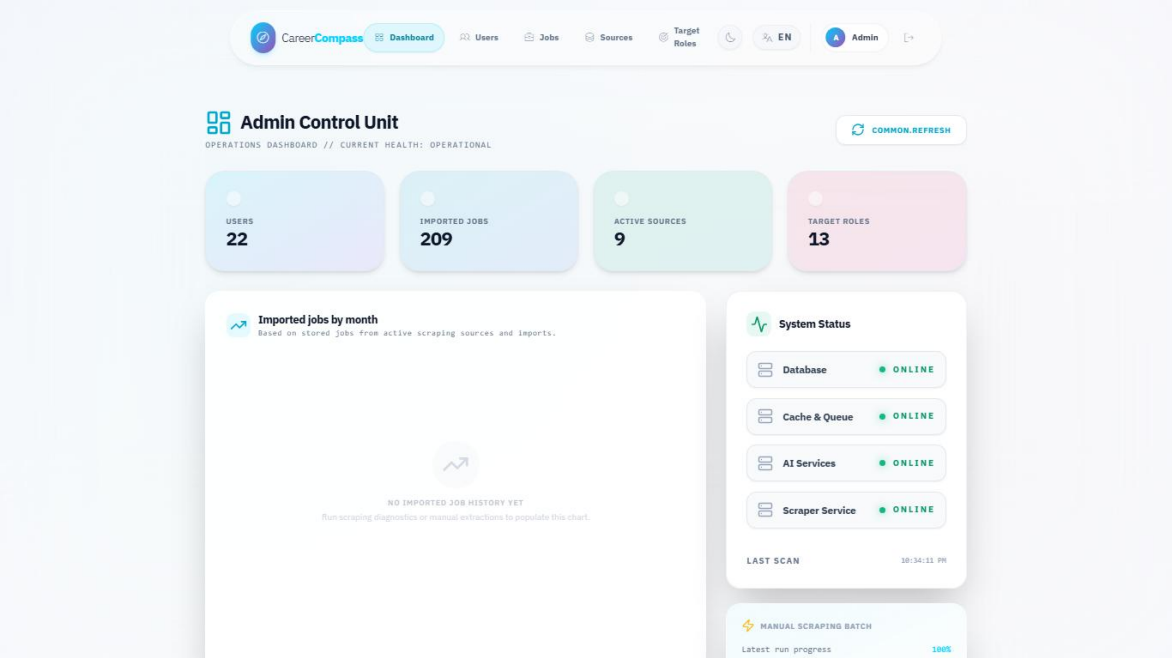

Figure 22. Admin dashboard.

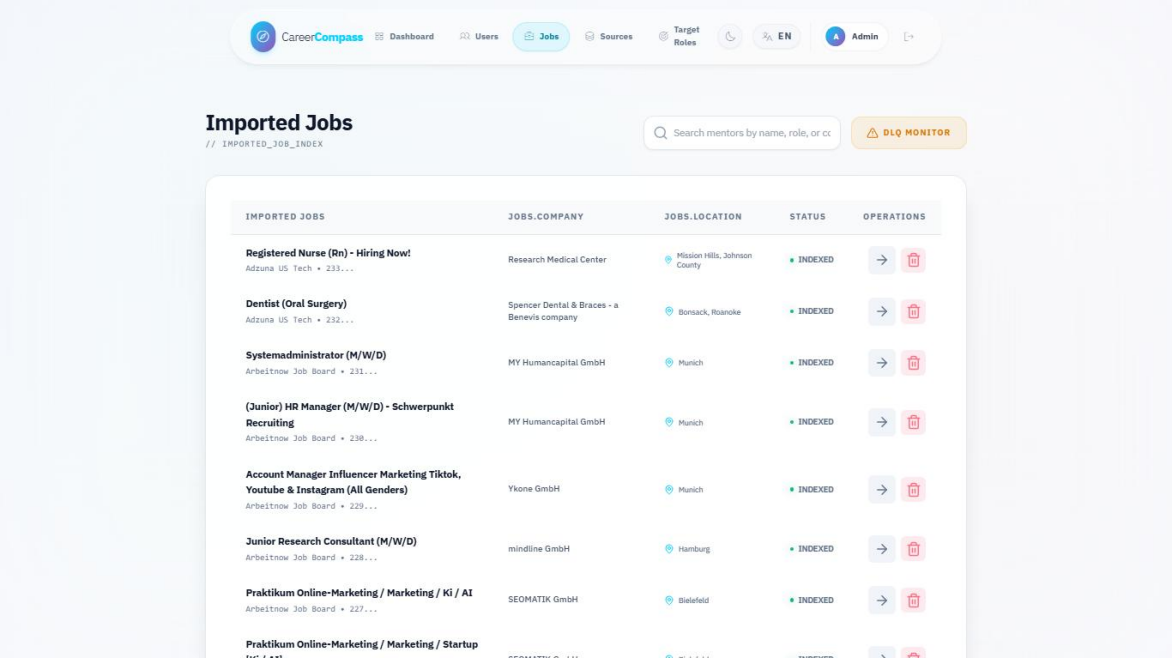

Figure 23. Admin jobs page.

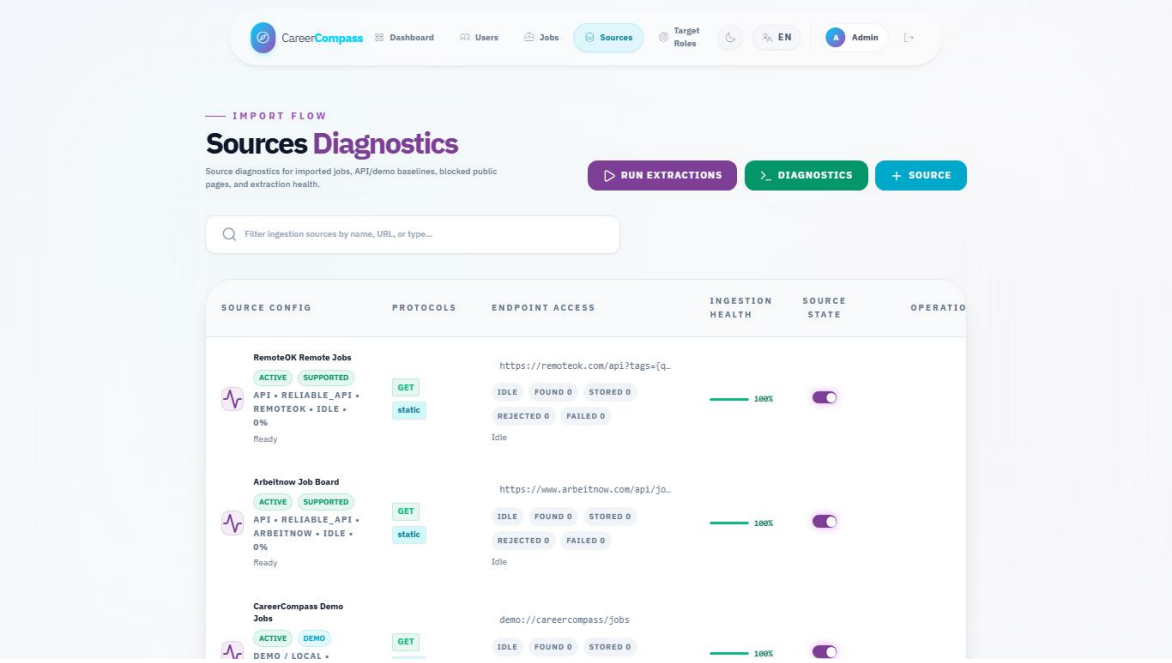

Figure 24. Admin sources diagnostics page.

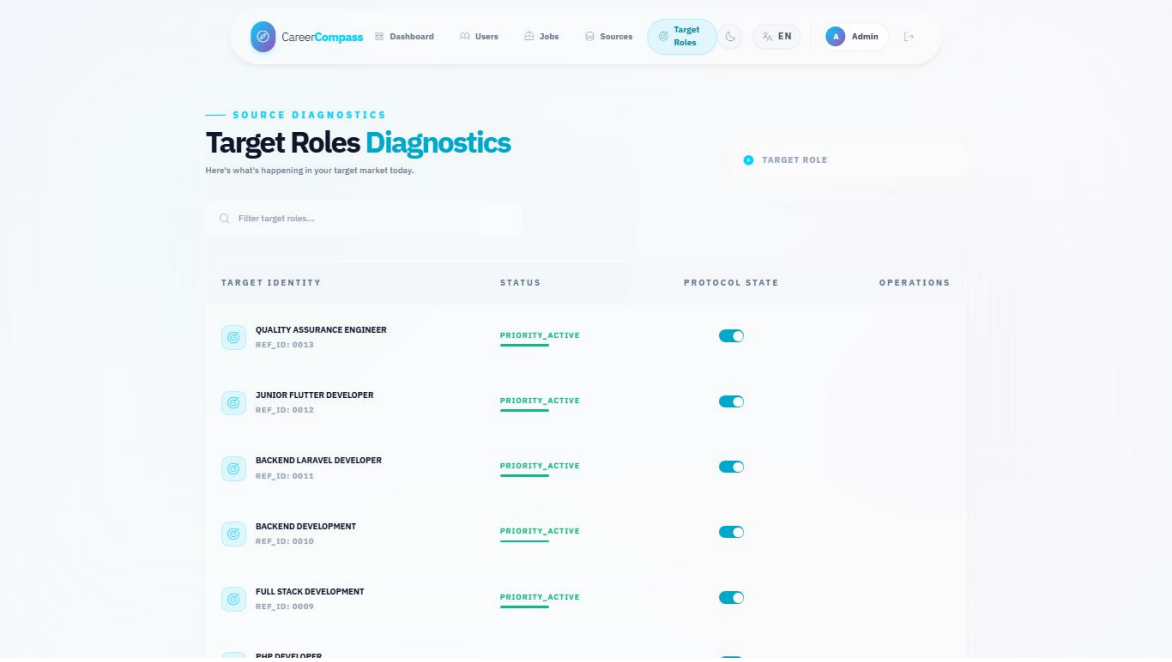

Figure 25. Admin target roles page.

Docker Compose Services Evidence

NAME	IMAGE	COMMAND	SERVICE	CREATED
cc-backend	graduation-project-backend-api	"/usr/local/bin/cc-eâ€¦"	backend-api	18 hours ago
cc-backend-scheduler	graduation-project-backend-scheduler	"/usr/local/bin/cc-eâ€¦"	backend-scheduler	18 hours ago
cc-backend-worker	graduation-project-backend-worker	"/usr/local/bin/cc-eâ€¦"	backend-worker	18 hours ago
cc-backend-worker-ai	graduation-project-backend-worker-ai	"/usr/local/bin/cc-eâ€¦"	backend-worker-ai	18 hours ago
cc-backend-worker-emails	graduation-project-backend-worker-emails	"/usr/local/bin/cc-eâ€¦"	backend-worker-emails	18 hours ago
cc-backend-worker-high	graduation-project-backend-worker-high	"/usr/local/bin/cc-eâ€¦"	backend-worker-high	18 hours ago
cc-backend-worker-scraping	graduation-project-backend-worker-scraping	"/usr/local/bin/cc-eâ€¦"	backend-worker-scraping	18 hours ago
cc-cv-analyzer	graduation-project-ai-cv-analyzer	"uvicorn main:app --â€¦"	ai-cv-analyzer	18 hours ago
cc-db	mysql:8.0	"docker-entrypoint.sâ€¦"	db	2 weeks ago
cc-frontend	graduation-project-frontend	"/docker-entrypoint.â€¦"	frontend	18 hours ago
cc-job-miner	graduation-project-ai-job-miner	"uvicorn service_apiâ€¦"	ai-job-miner	18 hours ago
cc-nginx	nginx:stable-alpine	"/docker-entrypoint.â€¦"	nginx	6 days ago
graduation-project-grafana-1	grafana/grafana-oss:11.4.0	"/run.sh"	grafana	2 weeks ago
graduation-project-minio-1	minio/minio:latest	"/usr/bin/docker-entâ€¦"	minio	2 weeks ago
graduation-project-prometheus-1	prom/prometheus:v2.55.1	"/bin/sh -c 'sed \"s/â€¦"	prometheus	2 weeks ago

Figure 26. Docker services evidence.

Validation Command Evidence

```
Validation summary captured for the graduation book:

docker compose config --quiet: PASSED
docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet: PASSED
docker compose up -d --build: stack built; full initial build exceeded 15 minutes, then targeted rebuild/start completed
composer install in backend-api container: PASSED
php artisan config:clear: PASSED
php artisan route:list: PASSED (131 routes)
php artisan migrate --force --no-interaction: PASSED (nothing to migrate)
php artisan test: PASSED (39 tests, 297 assertions)
frontend ESLint: PASSED with 9 warnings and 0 errors
frontend Vite build: PASSED (2904 modules transformed)
ai-job-miner pytest: PASSED (75 passed)
ai-cv-analyzer compileall: PASSED
ai-job-miner compileall: PASSED
ai-cv-analyzer pytest: SKIPPED/blocked because pytest is not installed in that container
HTTP probes: /, /api/health, /api/ready, /status, AI CV Analyzer root, and Job Miner health returned 200
```

Figure 27. Validation command evidence.

Appendix E: Test Cases

The manual test matrix in Chapter 6 should be repeated before final submission. Additional recommended tests include invalid CV uploads, banned user login, expired signed download URLs, failed AI service behavior, scraper token rejection, admin route rejection for normal users, and browser checks on a clean database.

The mini dataset evaluation files are:

- evaluation/mini_cv_dataset.json
- evaluation/mini_jobs_dataset.json
- evaluation/expected_labels.json
- evaluation/run_mini_evaluation.py
- evaluation/mini_evaluation_results.json
- evaluation/mini_evaluation_summary.md

Appendix F: GitHub Actions / CI Summary

The repository contains GitHub Actions workflow definitions. After the draft PR is opened, reviewers should inspect PR checks, rerun failed jobs if needed, and confirm that branch protection expectations are satisfied before merging. A CI screenshot was not embedded in this generated report because live PR checks were not available until after branch push.

Appendix G: Demo Script

1. Start Docker Desktop.
2. Run docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build.
3. Open http://localhost.
4. Register or login as a student.
5. Upload a sample PDF CV from the dashboard.
6. Review parsed profile and skills.
7. Open Jobs and select a recommendation.
8. Open Gap Analysis and explain matched/missing skills.
9. Save the job to Applications.
10. Login as the demo admin.
11. Open Admin Dashboard, Jobs, Sources, and Target Roles.
12. Open System Status and explain health checks.
13. Discuss limitations and future work honestly.